



Monitoring and Assessment of Emission and Dispersion of Gaseous Pollutants from Major Landfill Sites in Dhaka, Bangladesh

Intehum Taufique Aurnab ·
Shamsunnahar Khanam

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Abstract Landfills can be a major source of air pollution as anaerobic decomposition of organic waste in landfills releases gaseous pollutants like methane (CH_4), carbon dioxide (CO_2), and other non-methane organic compounds (NMOCs) in the atmosphere. However, the emission rates of these gases from Dhaka's two biggest landfills, namely Matuail and Aminbazar Landfills, have not yet been calculated, nor has their atmospheric dispersion been evaluated. This study therefore aimed to investigate the emission rates of different gases from Matuail and Aminbazar landfills located in Dhaka utilizing four simulations (CAAC, IC, IW, SP-1) of LandGEM V-3.02 model. The study also determined the distribution pattern of the emitted gases from Matuail and Aminbazar landfills in the atmosphere of Dhaka using AERMOD View software model. Finally, the study carried out field measurements surrounding Matuail and Aminbazar landfills using Haz-Scanner-TM model HIM-600. Results indicated that the emission rates of different gases from Matuail and Aminbazar landfills followed the order of $\text{CO}_2 > \text{CH}_4 > \text{NMOCs}$. Results of dispersion modeling showed that maximum concentrations of all the

gases were found near the landfills sites. However, the maximum concentrations of CH_4 and CO_2 near the landfill sites were within the allowable permissible limits established by United States National Institute for Occupational Safety and Health (NIOSH). According to the field measurements conducted surrounding both Matuail and Aminbazar landfills, average concentrations of CH_4 and CO_2 near Matuail landfill were 167.55 ppm and 534.74 ppm respectively while average concentrations of CH_4 and CO_2 near Aminbazar landfill were 182.1 ppm and 593.04 ppm respectively. Dispersion modeling based on emission data from IW simulation provided concentrations (Matuail: CH_4 —109.29 ppm, CO_2 —109.39 ppm, Aminbazar: CH_4 —159.78 ppm, CO_2 —166.87 ppm) closest to the findings obtained by field measurements conducted near both Matuail and Aminbazar landfills. The results of this study should aid in the development of efficient management plans and policy interventions to reduce the risks of air pollution caused by landfill operations in rapidly urbanizing cities.

Keywords LandGEM · AERMOD · Carbon dioxide · Methane · Non-methane organic compounds

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I. T. Aurnab (✉) · S. Khanam
Department of Environmental Science, Bangladesh
University of Professionals, Dhaka 1216, Bangladesh
e-mail: aurnab@bup.edu.bd

1 Introduction

The increase in population and urbanization has led to the generation of substantial quantity of industrial

and municipal solid waste worldwide (Talaiekhazani et al., 2018a, b; Tisserant et al., 2017). International Solid waste Association (ISWA) reports that there is a global generation of 2.6 million tons/day of MSW, which is projected to increase to 4.5 million tons/day by the year 2050 (ISWA, 2021; Salami & Popoola, 2023). Proper disposal of the substantial volume of waste produced globally is imperative and land-filling is a commonly employed approach of MSW disposal specially in developing nations (Elmi et al., 2021; Ferronato & Torretta, 2019; Gupta et al., 2015; Salami & Popoola, 2023).

After being disposed of in the landfills, the organic elements in the waste can undergo chemical and biological interactions that lead to the degradation of these substances. The process of decomposition produces a substantial quantity of gases, which are subsequently emitted into the air. The gases emitted from landfills are commonly known as biogas or landfill gas. The main constituents of biogas are carbon dioxide (CO_2), methane (CH_4), and non-methane volatile organic compounds (NMOCs) (Omar & Rohani, 2017; Talaiekhazani et al., 2018a). Biogas consists of around 60% CH_4 , 40% CO_2 , and less than 1% NMOCs (Talaiekhazani et al., 2018b). Hence, inadequate landfill management can lead to release of different air pollutants into the atmosphere, thereby polluting the atmosphere and deteriorating air quality (Talaiekhazani et al., 2018b).

Bangladesh, similar to other developing nations, has overlooked its solid waste management program. The quantity of solid waste in Dhaka doubled between 2014 and 2021, which is a cause for concern (DNCC, 2017, 2019; DSCC, 2020; Toha & Rahman, 2023). In Bangladesh, a substantial component of MSW is composed of organic materials (68–81%) that undergo anaerobic decomposition (Shams et al., 2017). Landfills in Dhaka produce a large amount of landfill gas (LFG) on a daily basis due to the high concentration of organic waste. Methane makes up approximately 55–60% of the total volume of LFG (Zhan et al., 2020). Consequently, the emission of gaseous pollutants from Dhaka's landfills i.e. Matuail and Aminbazar landfills can have the potential to pose threat to environment as well as human health (Salami & Popoola, 2023; Toha & Rahman, 2023), and to ensure proper management and reduce gaseous emissions from the landfills it is necessary to estimate the rate at which gaseous pollutants are emitted

from Dhaka's landfills and determine how the emitted gases are distributed in Dhaka's atmosphere.

Multiple techniques can be employed to assess gaseous emissions from landfills, such as site assessment, experimental testing in the field, and the use of mathematical models (Knapp et al., 2014; Talaiekhazani et al., 2018a). The LandGEM model is a tool that is fully automated which allows for the calculation of yearly emissions from landfills utilizing a first-order decomposition rate equation (Bruce et al., 2017; Yadollahikhales et al., 2016). US Environmental Protection Agency (USEPA) has designed this model (Talaiekhazani et al., 2018a). LandGEM is a tool that calculates and predicts the release of CO_2 , CH_4 , and NMOCs from landfills (Bruce et al., 2017; Saqur et al., 2007; Yadollahikhales et al., 2016). AERMOD View is a model that is used to understand the spread of air pollutants over the specified area. AERMOD, developed by American Meteorological society (AMS) and United States Environmental Protection Agency (US EPA) is based on Gaussian plume model (Elmi et al., 2021). This software possesses the ability to model the distribution of atmospheric contaminants originating from both point and area sources (Asadi et al., 2017; Dey et al., 2017; Esbrí et al., 2015).

Previous studies have highlighted that substantial quantity of gases including CH_4 , CO_2 , and NMOCs have been released from different landfills around the world, particularly in developing countries and these emitted gases from the landfills were dispersed into the atmosphere (Elmi et al., 2021; Talaiekhazani & Nasiri, 2016; Talaiekhazani et al., 2016, 2018a, b; Wijaya et al., 2021). Elmi et al. (2021) modeled emission and dispersion of CH_4 and CO_2 from landfill sites located in Kuwait. Talaiekhazani and Nasiri (2016), Talaiekhazani et al. (2016), Talaiekhazani et al. (2018a), and Talaiekhazani et al. (2018b) estimated emission and dispersion of CH_4 , CO_2 , and NMOCs from different landfills sites located in Iran. Wijaya et al. (2021) estimated CH_4 emission and dispersion from Sarimukti landfill located in Indonesia. However, no study has estimated emission rate of gaseous pollutants like CH_4 , CO_2 , and NMOCs from Aminbazar and Matuail landfills located in Dhaka and determined the dispersion pattern of the emitted gases in the atmosphere of Dhaka based on the emission data. This study utilized the LandGEM V-3.02 model to calculate the emission of gases from Matuail and

Aminbazar landfills and applied the AERMOD View software model to simulate the dispersion pattern of the emitted gases from Matuail and Aminbazar landfills. This study also carried out field measurements to compare the findings of the field measurements with the model outputs. The objectives of this study are:

- i. to estimate the rate of emission of gaseous pollutants including methane (CH_4), carbon dioxide (CO_2), and non-methane organic compounds (NMOCs) from Matuail and Aminbazar landfill sites in Dhaka,
- ii. to simulate the dispersion pattern of the emitted gases from the landfills in the atmosphere of Dhaka,
- iii. to conduct field measurements for determining the concentrations of air pollutants like methane (CH_4), carbon dioxide (CO_2), carbon monoxide (CO), nitric oxide (NO), nitrogen dioxide (NO_2), sulfur dioxide (SO_2), ozone (O_3), PM_{10} , and $\text{PM}_{2.5}$ surrounding Matuail and Aminbazar landfill sites,
- iv. to compare findings obtained by field measurements with the model outputs.

Information and data obtained in this study would be essential for developing effective management strategies to reduce air pollution from landfill sites in Dhaka.

2 Methodology

2.1 Study Area

The areas selected for this study are Matuail and Aminbazar landfills which are located between $23^\circ 43' 2.44''\text{N}$ latitude and $90^\circ 27' 4.53''\text{E}$ longitude and $23^\circ 47' 52.75''\text{N}$ latitude and $90^\circ 18' 00.85''\text{E}$ longitude respectively (Fig. 1). The management of municipal solid waste (MSW) in Dhaka is primarily overseen by two municipal corporations which are Dhaka South City Corporation (DSCC) and Dhaka North City Corporation (DNCC). Matuail landfill is under DSCC while Aminbazar landfill is administered by DNCC (Toha & Rahman, 2023).

The Matuail landfill, managed by the DSCC, is located approximately 6 km south of Dhaka, in the Matuail Union. It was initially built in 1995 as an open dumping site with an area of 50 acres (20 ha).

In 2006, an additional 50 acres (20 ha) were added to the landfill. Municipal solid waste (MSW) of 57 wards under five zones of DSCC is dumped in Matuail landfill (Urme et al., 2021). The height of the municipal solid waste (MSW) piles in Matuail landfill is 21 m (Toha & Rahman, 2023) (Table A.1).

The Aminbazar landfill, located in Savar upazila, 8 km northwest of Dhaka city, was established in 2007 on a 52-acre (21 hectare) plot of land under the DNCC. This landfill covers nearly five zones and 36 wards in the DNCC (Urme et al., 2021). The height of the municipal solid waste (MSW) piles in Matuail landfill is 18 m (Toha & Rahman, 2023) (Table A.1).

3 Research Design

This research study is exclusively anchored on software modeling and field measurement (Fig. 2). The current study employs a quantitative research methodology. Firstly, description of the study area (Aminbazar and Matuail landfills) was provided based on information collected from published literatures. Then, description of the waste of Dhaka's landfills including the waste generation rate as well as waste composition was provided based on the data collected from waste reports of DSCC and DNCC. After that climatic conditions of Dhaka was determined. Then emissions of total landfill gas, methane (CH_4), carbon dioxide (CO_2), and non-methane organic compounds (NMOCs) from both Matuail and Aminbazar landfills were estimated using LandGEM V-3.02 model. Then dispersion of total landfill gas, methane (CH_4), carbon dioxide (CO_2), and non-methane organic compounds (NMOCs) from both Matuail and Aminbazar landfills were modeled using AERMOD software. Then field measurement was carried out in different sampling points surrounding both Aminbazar and Matuail landfills and concentrations of different air pollutants like methane (CH_4), carbon dioxide (CO_2), carbon monoxide (CO), nitric oxide (NO), nitrogen dioxide (NO_2), sulfur dioxide (SO_2), ozone (O_3), PM_{10} , and $\text{PM}_{2.5}$ were monitored using Haz-Scanner-TM model HIM-600. Finally, Statistical analysis was conducted.

3.1 Quantity and Composition of Waste of Dhaka's Landfills

The waste generation rate from Matuail and Aminbazar landfills was 687.466 Gg/year and 683.174 Gg/

year respectively in 2015. From 2015 to 2021, the waste generation rate of both Matuail and Aminbazar landfills showed an increasing trend. In 2021, the waste generation rate in Matuail landfill was 1224.72 Gg/year which was 1.78 times higher as compared to the waste generation rate of 2015. Similarly, the waste generation rate of Aminbazar landfill in 2021 (1211.776 Gg/year) was 1.77 times higher as compared to 2015 (DNCC, 2017, 2019; DSCC, 2020) (Fig. B.1).

The solid wastes of Dhaka's landfills include 67.65% fruits and vegetables, 12.23% paper and rags, 5% metals, 4.20% wood, 3% plastic, 2.50% leaves, hay, and straw, 1% glass, 1% leather and rubber, 0.50% stone and rocks, and 2.92% other miscellaneous (Toha & Rahman, 2023). The composition of the waste of Dhaka's landfills is shown in Fig. B.1.

3.2 Assessment of the Climatic Conditions of Dhaka

Weather conditions are among the most significant environmental factors that are taken into consideration when managing landfills (Talaiekhosani et al., 2018b). It is necessary to determine a portion of the input data required for estimating the gaseous emissions from Dhaka's landfills using the LandGEM model based on the climate of Dhaka. For this investigation, Eq. (1), often known as the De Martonne equation, was applied to identify the climatic conditions of Dhaka. This equation was used to calculate the regional dryness index (I), which was accomplished by providing the average annual rainfall and mean annual temperature of Dhaka. Following that, the climate of Dhaka was determined

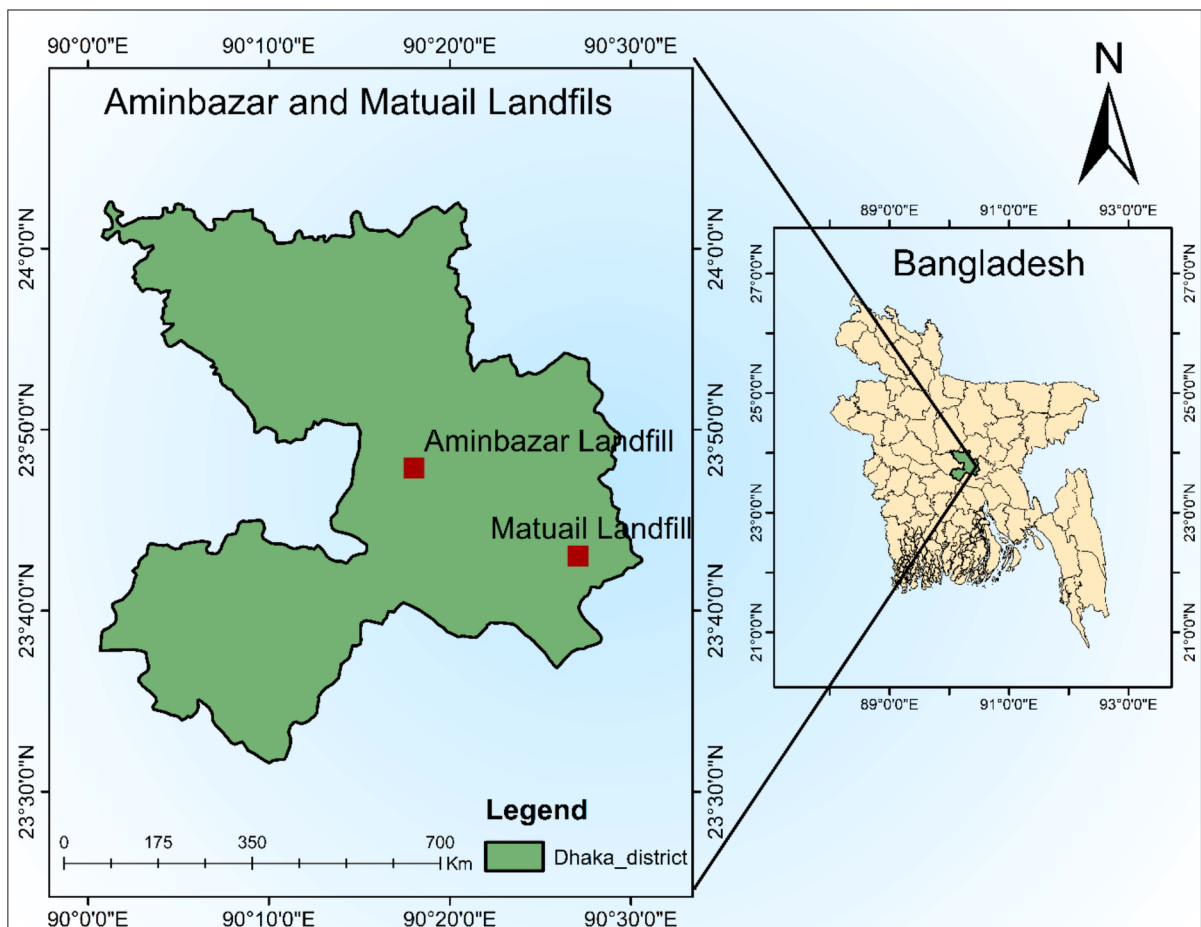


Fig. 1 Figure showing the study area (Aminbazar and Matuail landfills)

by using Table 1 in accordance with the findings of the calculation (Talaiekhazani et al., 2018a, b). The equation for calculating the regional dryness index (I) is as follows:

$$I = P/(T + 1) \quad (1)$$

In Eq. (1), I=regional dryness index, P=annual average precipitation in mm, T=annual average temperature in °C. The annual average precipitation and annual average temperature of Dhaka are 2076 mm and 25 °C respectively (Toha & Rahman, 2023).

3.3 Estimation of the Emissions of Gaseous Pollutants

Landfill Gas Emissions Model (Land-GEM) V-3.02 developed by US Environmental Protection Agency (US EPA) was utilized to estimate and predict the emissions of CO₂, CH₄, and Non-methane organic compounds from Aminbazar and Matuail landfills. LandGEM has been used extensively to estimate global landfill gas emissions (Talaiekhazani et al., 2018a, b; Toha and Rahman, 2023). Eq. (2) was used by LandGEM model to estimate the emission rate of CH₄ from landfills.

$$Q_{CH_4} = \sum_{i=1}^n \sum_{j=0.1}^1 kL_0 \left(\frac{M_i}{10} \right) e^{-kt_{ij}} \quad (2)$$

In Eq. (2), Q_{CH_4} = annual methane generation in the year of the calculation (m³/year), i=the 1-year time increment, n=(year of the calculation)—(initial year of waste acceptance), j=0.1-year time increment, k=the methane generation rate (year⁻¹), L_0 =potential methane generation capacity (m³/ton); M_i =mass of wastes accepted in the *i*th year (tons), t_{ij} =age of the *j*th section of waste mass M_i accepted in the *i*th year (in tenths of a year).

The methane generation rate (K) represents the rate at which methane is produced in relation to the mass of solid waste in the landfill. A larger value of k corresponds to an accelerated rate of methane generation. The primary determinant of the value of k is the level of moisture in waste. The landfills can be categorized into three groups which are dried landfills, conventional landfills and wet landfills. Based

on the landfill type, LandGEM model has different simulations including CAA Conventional (CAAC), CAA Arid Area (CAAA), Inventory Conventional (IC), Inventory Arid Area (IA), Inventory Wet (IW), each with different methane generation rate (K) and different potential methane generation capacity (L_0) values (Alexander et al., 2005; Talaiekhazani et al., 2018a, b; Toha & Rahman, 2023). Arid area landfills are situated in regions with an annual rainfall of less than 635 mm (Talaiekhazani et al., 2018a). The annual average precipitation of Dhaka is 2076 mm (Toha & Rahman, 2023). Hence, in this study CAA Conventional (CAAC), Inventory Conventional (IC), and Inventory Wet (IW) have been selected and CAA Arid Area (CAAA) and Inventory Arid Area (IA) have been excluded. Additionally, site specific-1 (SP-1) validated by Toha & Rahman (2023) has also been utilized in this study (Table 2).

Potential methane generation capacity (L_0) is another coefficient utilized in the LandGEM model that is influenced by the specific type and mix of solid waste that is deposited in the landfill. A higher concentration of cellulose results in an elevated L_0 (Alexander et al., 2005).

The coefficient of NMOC concentration in landfill gas is determined by the mix of solid waste in the landfill and the level of reactions that generate different chemicals through the anaerobic decomposition of waste. The NMOC concentration coefficient is calculated in parts per million by volume (ppmv) and is utilized exclusively by LandGEM for estimating NMOC emissions. The LandGEM model considered that landfill gas consists of an equal mixture of 50% methane and 50% carbon dioxide, along with small amounts of non-methane organic compounds (NMOCs) and other air pollutants (Alexander et al.,

Table 1 Classifications of climate based on regional dryness index

Range of Values	Classifications of Climate
$I < 10$	Dry
$10 \leq I \leq 19.9$	Semi dry
$20 \leq I \leq 23.4$	Mediterranean
$24 \leq I \leq 27.9$	Semi humid
$28 \leq I \leq 34.9$	Humid
$I > 35$	Intense humidity

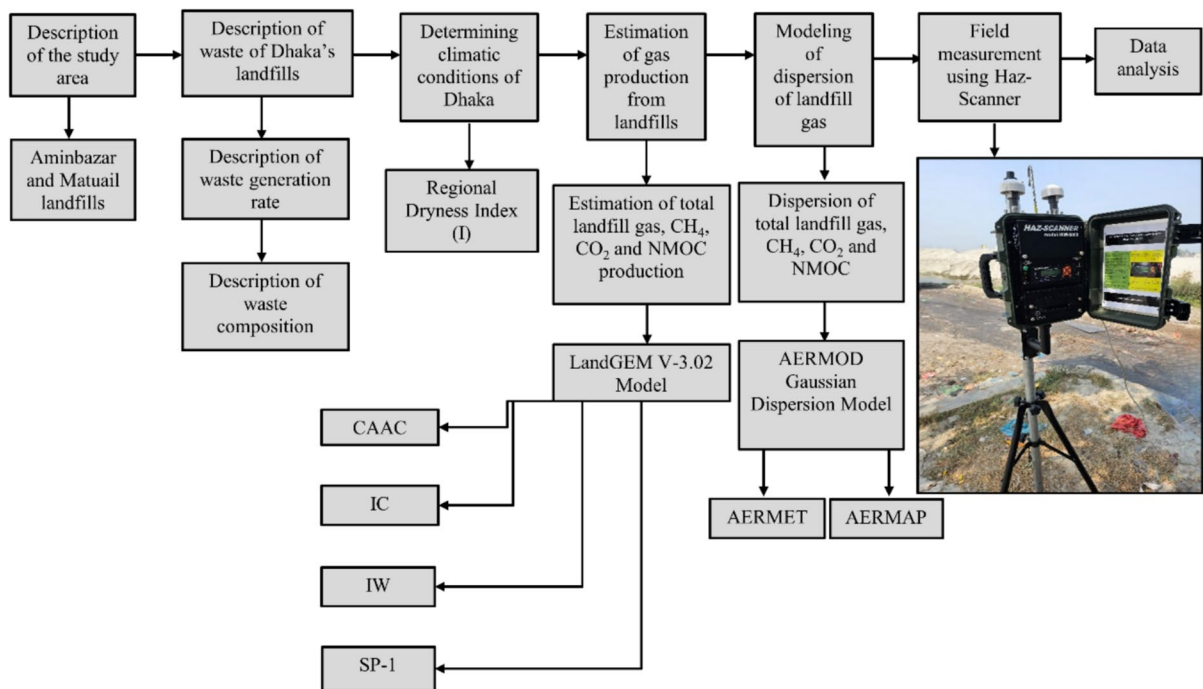


Fig. 2 Flow diagram showing the research design of the study

2005; Talaiekhosani et al., 2018a, b; Toha & Rahman, 2023).

For calculation of the emission rate of carbon dioxide (CO_2) from the estimated emission rate of methane (CH_4), Eq. (3) was applied. The equation used for the estimation of CO_2 emissions from Aminbazar and Matuail landfills is as follows:

$$Q_{\text{CO}_2} = Q_{\text{CH}_4} \times \left\{ \left[1 / \left(P_{\text{CH}_4} / 100 \right) \right] - 1 \right\} \quad (3)$$

In Eq. (3), Q_{CH_4} = annual methane generation in the year of the calculation (m^3/year), Q_{CO_2} = annual carbon dioxide generation in the year of the calculation (m^3/year), and P_{CH_4} = methane content percentage.

3.4 Uncertainty Analysis

The accuracy of estimating landfill gas generation using LandGEM model at any location is significantly influenced by the input parameters (Table 2). The input parameters are methane generation rate (K), potential methane generation capacity (L_0), methane content, and NMOCs concentration. Determining these

parameters is often complex, leading individuals to rely on suggested or default values. For example, values for parameters like methane generation rate (K) can range from $0.02\text{--}0.7 \text{ year}^{-1}$, potential methane generation capacity (L_0) can range from $96\text{--}170 \text{ m}^3/\text{ton}$, methane content can range from $40\text{--}60\%$, and NMOCs concentration can range from $600\text{--}4000 \text{ ppmv}$ as hexane. These create uncertainties in estimating landfill gas emissions that require careful evaluation. The uncertainty analysis in this study was conducted following Chandra & Ganguly (2023) and Talaiekhosani et al. (2018a).

3.5 Modeling the Dispersion of Gaseous Pollutants

AERMOD (American Meteorology Society Environmental Protection Agency Regulatory Model) View software was used for simulating the dispersion pattern of total landfill gas, CH_4 , CO_2 , and NMOCs in 20 km radius around both Aminbazar and Matuail landfills (Fig. 3). AERMOD, developed by American Meteorological society (AMS) and United States Environmental Protection Agency (US EPA) is based on Gaussian plume model (Elmi et al., 2021) which

Table 2 Data used in four different simulations of LandGEM model (Toha & Rahman, 2023)

Parameter	Unit	CAAC	IC	IW	SP-1
K	year ⁻¹	0.05	0.04	0.7	0.07
L ₀	m ³ /ton	170	100	96	96
Concentration of NMOCs	ppmv as hexane	4000	4000	4000	4000
Methane components	% by volume	50	50	50	50

can simulate concentrations of different air pollutants emitted from both point and area sources over a range of 50 km (Asadi et al., 2017; Dey et al., 2017; Esbri et al., 2015).

The AERMOD modeling system consists of two regulatory components for processing input data: AERMET and AERMAP. AERMET is a meteorological data preprocessor that incorporates air dispersion based on planetary boundary layer turbulence structure and scaling concepts. AERMAP, on the other hand, is a terrain data preprocessor that incorporates complex terrain using USGS Digital Elevation Data (Talaiekhosani et al., b). For running AERMOD, hourly meteorological data of Dhaka including opaque cloud cover, dry bulb temperature, relative humidity, station pressure, wind direction, and wind speed were provided. The meteorological data used in the dispersion modeling were collected from Bangladesh Meteorological Department (BMD) for the period of January to December 2023. Table A.2 represents the hourly average yearly meteorological data of Dhaka for 2023. The dispersion modeling is based on numerous assumptions (Elmi et al., 2021). The conditions were assumed to be in a steady state. The background concentrations of CH₄, CO₂, and NMOCs were assumed to be zero. The emissions from other sources, such as transportation and industrial activities, were considered to be insignificant. However, these assumptions might not indicate the actual situation.

3.6 Field Measurement

For conducting field measurements, concentrations of different air pollutants including methane (CH₄), carbon dioxide (CO₂), carbon monoxide (CO), nitric oxide (NO), nitrogen dioxide (NO₂), sulfur dioxide (SO₂), ozone (O₃), PM1, and PM2.5 were monitored using Haz-Scanner-TM model HIM-600 randomly from six sampling points (three near Matuail landfill and three near Aminbazar

landfill) around Matuail and Aminbazar landfills. In case of both Matuail and Aminbazar landfill site, one sampling point was located close to the waste dumping area and the other two sampling points were placed in the nearby surrounding areas. Haz-scanner is equipped with particulate matter, temperature, humidity sensor to detect different variables (Sin Ko Ko et al., 2022). This equipment is also equipped with variety of gas sensors (Toha & Rahman, 2023) that extracts air and quantifies the concentrations of different air pollutants including methane (CH₄), carbon dioxide (CO₂), carbon monoxide (CO), nitric oxide (NO), nitrogen dioxide (NO₂), sulfur dioxide (SO₂), ozone (O₃),. The sensor technologies of a Haz-scanner vary depending on the pollutants being measured. For examples, light scattering technology is used for detection of PM1, PM2.5, metal oxide semiconductor (MOS) is used for O₃, electrochemical sensor (EC) is used for detection of gases like CO, NO_x, SO₂ etc. and non dispersive infrared (NDIR) technology is used for detection of gases like CH₄, CO₂ etc. The instrument provides direct readings in real time (HAZ-SCANNER EPAS, (HAZ-SCANNER n.d.)). Field measurement using Haz-Scanner-TM model HIM-600 was carried out for 1 h at 1 min interval at each sampling point following Toha & Rahman (2023) who conducted field measurements for 30 min at 5-min intervals,

3.7 Statistical Analysis

Pearson correlation (r) analysis was conducted between different simulations of LandGEM model (CAAC, IC, IW, SP-1). In Pearson correlation analysis, r=0.10–0.30 indicates low correlation, r=0.40 indicates medium correlation, and r=0.70 to 1 indicates high correlation and correlation is statistically significant when $p < 0.01$ (99% confidence level) or $p < 0.05$ (95% confidence level)

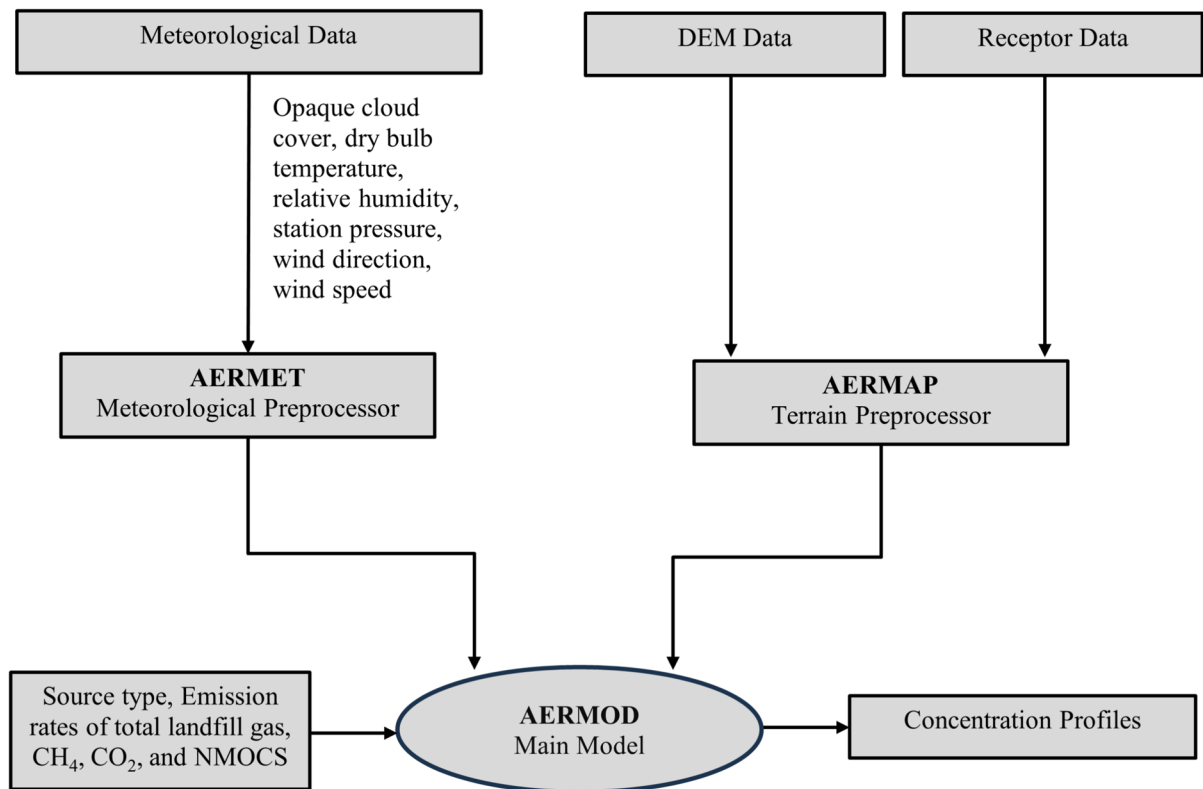


Fig. 3 Flow diagram showing the AERMOD modeling system

(Dancey & Reidy, 2004). In addition to Pearson correlation analysis, regression analysis was conducted to determine the link between methane generation rate constant (K) and peak total landfill gas, CH₄, CO₂, and NMOC generation values. Pearson correlation analysis was conducted using SPSS V-26 software package and regression analysis was performed using Microsoft Excel.

4 Results and Discussion

4.1 Climate of Dhaka

The annual average precipitation of Dhaka is 2076 mm while the annual average temperature of Dhaka is 25 °C (Toha & Rahman, 2023). Assessment of the climatic conditions of Dhaka provided a regional dryness index (I) value of 79.85, indicating that Dhaka's climate is intensely humid. Hourly average yearly opaque cloud cover, dry bulb temperature,

relative humidity, station pressure, wind direction, and wind speed of Dhaka for the year 2023 are 4 (tenths), 27 °C, 71%, 1009 mbar, 7 degrees and 1 m/s respectively, as presented in Table A.2.

A wind rose (Fig. 4) displays the predominant wind's speed and direction over Dhaka in 2023. Majority of the winds (54.75%) are classified as calm winds, with the remaining winds (45.25%) having a discernible speed. The wind rose clearly indicates that the prevailing wind direction is from northeast to southwest. Since Matuail landfill is located south of Dhaka city and Aminbazar landfill is located northwest of Dhaka city, it is less likely that the prevailing winds have the capability of transporting the released gaseous pollutants from landfills towards the city. According to (Ghiaseddin, 2015) and (Talaiekhazani et al., 2018a), a wind speed of 8.3 m/s is regarded sufficient to disperse the released pollutants. Nonetheless, Dhaka experiences a scarcity of effective winds, as depicted in Fig. 5.

4.2 Emission of Total Landfill Gas from Matuail and Aminbazar Landfills

The emission of total landfill gas, also known as biogas from both Matuail and Aminbazar landfills, estimated by four different simulations of LandGEM V-3.02 model (CAAC, IC, IW, SP-1) is shown in Fig. 5.

According to the output obtained by the model, CAAC, IC, IW, and SP-1 predicted that 14.27 Gg/year, 6.75 Gg/year, 85.92 Gg/year, and 11.18 Gg/year of biogas respectively had been emitted from Matuail landfill in 2016. On the other hand, results indicated that Aminbazar landfill showed less emission rates as compared to Matuail landfill in 2016. In case of Aminbazar landfill the amount of biogas emission in 2016 predicted by CAAC, IC, IW, and SP-1 are 14.18 Gg/year, 6.70 Gg/year, 85.38 Gg/year, and 11.11 Gg/year respectively. However, as years showed by, Aminbazar landfill showed higher emission rates compared to Matuail landfill which is similar to the findings of (Toha & Rahman, 2023). Biogas emission rate reached its peak in 2022 in case of Matuail landfill and in 2023 in case of Aminbazar landfill. The biogas emission rate from Matuail landfill in 2022 predicted by CAAC, IC, IW, and SP-1 are 116.33 Gg/year, 56.35 Gg/year,

273.43 Gg/year, and 86.91 Gg/year respectively. On the other hand, CAAC, IC, IW, and SP-1 predicted that highest amount of biogas had been emitted from Aminbazar landfill in 2023 which are 150.82 Gg/year, 73.41 Gg/year, 295.57 Gg/year, and 111.60 Gg/year respectively. The peak biogas emission rates predicted by four simulations of LandGEM models were in the order of IW > CAAC > SP-1 > IC. After the peak biogas emission reached in 2022 and 2023 respectively for Matuail and Aminbazar landfills, the emission rate will continue to decrease till 2155. The peak biogas emission rates from Aminbazar and Matuail landfills were found to be greater than some other landfills around the world such as Borujerd landfill (Talaiekhosani et al., 2018a), Shahrekord landfill (Talaiekhosani et al., 2018b), Akortiri landfill (Chalvatzaki & Lazaridis, 2010). A probable reason for this disparity is the substantially higher waste input at Dhaka's landfill sites, along with a higher percentage of biodegradable organic matter in the deposited garbage (Fig. B.1). The quantity of rapidly degradable organic materials stimulates microbial activity, resulting in a more intensive biogas generating process (Toha & Rahman, 2023).

Regression analysis of methane generation constant (K) and peak landfill gas generation revealed

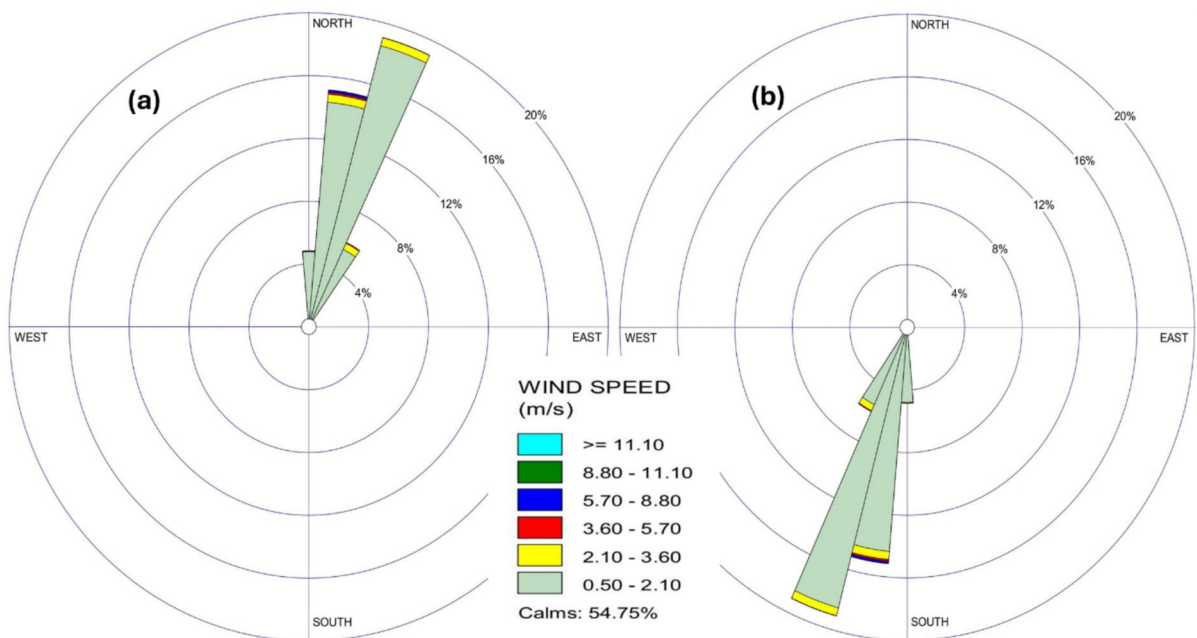


Fig. 4 Wind rose of Dhaka for 2023 showing the directions from which wind is (a) blowing from and (b) blowing to

that, in case of Matuail landfill, the equation is $y = 289.58x + 70.994$, $R^2 = 0.9407$ (Fig. B.4). It implies that 94.07% of the variability in the peak value (dependent variable) should be accounted for by the variability in the methane production constant (K) value. For Aminbazar landfill, the equation is $y = 284.73x + 96.632$, $R^2 = 0.9003$ (Fig. B.4), which indicates that 90.03% of the variability in the peak value (dependent variable) should be accounted for by the variability in the methane production constant (K) value.

Pearson correlation analysis between different simulations of LandGEM model (CAAC, IC, IW, SP-1) revealed that in case of Matuail landfill there is strong significant correlation between CAAC and IC at 99% confidence level ($r = 0.995$, $p < 0.01$), moderate significant correlation between CAAC and IW at 99% confidence level ($r = 0.532$, $p < 0.01$), strong significant correlation between CAAC and SP-1 at 99% confidence level ($r = 0.987$, $p < 0.01$), moderate significant correlation between IC and IW at 99% confidence level ($r = 0.476$, $p < 0.01$), strong significant correlation between IC and SP-1 at 99% confidence level ($r = 0.964$, $p < 0.01$), and moderate significant correlation between IW and SP-1 at 99% confidence level ($r = 0.619$, $p < 0.01$) (Table A3). Similar results were found in case of Aminbazar landfill. For Aminbazar landfill, Pearson correlation analysis revealed that there is strong significant correlation between CAAC and IC at 99% confidence level

($r = 0.994$, $p < 0.01$), moderate significant correlation between CAAC and IW at 99% confidence level ($r = 0.538$, $p < 0.01$), strong significant correlation between CAAC and SP-1 at 99% confidence level ($r = 0.986$, $p < 0.01$), moderate significant correlation between IC and IW at 99% confidence level ($r = 0.481$, $p < 0.01$), strong significant correlation between IC and SP-1 at 99% confidence level ($r = 0.964$, $p < 0.01$), and moderate significant correlation between IW and SP-1 at 99% confidence level ($r = 0.627$, $p < 0.01$) (Table A.4). The correlation analysis (Table A.3 and A.4) indicates that within the four LandGEM simulations, the CAAC and IC models show the highest correlations ($r = 0.995$ for Matuail and $r = 0.994$ for Aminbazar, $p < 0.01$). This indicates that both models relied on almost identical input parameters (Table 2). Toha and Rahman (2023) also reported high correlations between CAAC and IC.

4.3 Uncertainty in Landfill Gas Emissions

The LandGEM model results can be affected by uncertainty arising from assumptions and variability in input parameters such as methane generation rate (K), potential methane generation capacity (L_0), methane content, and NMOCs concentration. The results of uncertainty analysis for Matuail Landfill (Fig. 6) revealed that methane generation rate (K) influenced the generation of total landfill gas methane (CH_4), carbon dioxide (CO_2), and NMOCs in a similar way,

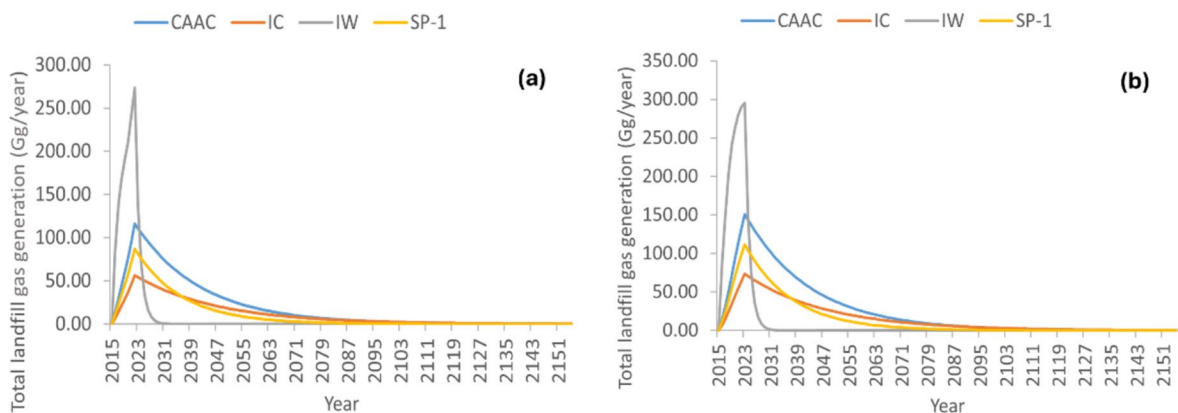


Fig. 5 Graph showing total landfill gas generation rate predicted by four simulations of LandGEM V-3.02 model from (a) Matuail landfill and (b) Aminbazar landfill

where an increase in K value from 0.02–0.7 year⁻¹ led to increased gas production. potential methane generation capacity (L_0) also had the same influence on landfill gas generation as methane generation rate (K). Increase in L_0 value from 96–170 m³ also led to increased gas production. Increase in methane content from 40 to 60% decreased the total landfill gas, CO₂, and NMOCs generation, while having no effect on CH₄ generation. Increase in NMOCs concentration from 600–4000 ppmv as hexane affected only NMOC generation, leading to a corresponding rise in its production. These findings are consistent with those of Talaiekhosani et al. (2018a) which also reported significant variations in landfill gas emissions caused by uncertainties in the values of K, L_0 , methane content and NMOCs concentration. The uncertainty analysis conducted for Aminbazar landfill (Fig. B.8) also provided similar results to Matuail landfill.

4.4 Dispersion of Total Landfill Gas from Matuail and Aminbazar Landfills

Results of biogas dispersion from Matuail landfill in 2023 using AERMOD View software based on four simulations of LandGEM model i.e. CAAC, IC, IW, and SP-1 is shown in Fig. 7. Results revealed that highest concentrations of biogas were found surrounding the Matuail landfill. Talaiekhosani et al. (2018a), Talaiekhosani et al. (2018b), and Wijaya et al. (2021) also found maximum concentrations of biogas surrounding the landfill sites. According to Talaiekhosani & Nasiri (2016) maximum pollutant concentration is found within 200 m of the landfill site. Based on the predicted total landfill gas emission rate by CAAC, IC, IW, and SP-1, biogas concentrations surrounding the Matuail landfill ranged from 200,100 µg/m³–218,872 µg/m³, 100,000 µg/m³–107,038 µg/m³, 200,600 µg/m³–268,605 µg/m³, and 100,600 µg/m³–160,304 µg/m³ respectively. Maximum biogas concentration surrounding Matuail landfill based on four different simulations of LandGEM model followed the order of IW > CAAC > SP-1 > IC. Biogas concentrations in most areas of Dhaka due to emissions from Matuail landfill were estimated to be in the ranges of 9100 µg/m³–90,100 µg/m³, 8000 µg/m³–80,000 µg/m³, 9600 µg/m³–90,600 µg/m³, and 8600 µg/m³–80,600 µg/m³ based on emission data from CAAC, IC, IW, and SP-1 respectively. (Talaiekhosani et al., 2018a)(Talaiekhosani et al., 2018b).

Maximum biogas concentrations surrounding the Matuail landfill were found to be higher than some other landfills around the world such as Borujerd landfill (Talaiekhosani et al., 2018a) and Shahrekord landfill (Talaiekhosani et al., 2018b) (Table 3).

Figure 8 shows Results of biogas dispersion from Aminbazar landfill in 2023 using AERMOD View software based on four simulations of LandGEM model i.e. CAAC, IC, IW, and SP-1. Maximum concentrations of biogas were estimated surrounding the Aminbazar landfill which is similar to the findings of Matuail landfill and other studies on different landfills around the world such as as Borujerd landfill (Talaiekhosani et al., 2018a), Shahrekord landfill (Talaiekhosani et al., 2018b), and Sarimukti landfill (Wijaya et al., 2021). Maximum biogas concentrations surrounding Aminbazar landfill were in the ranges of 200,000 µg/m³–209,430 µg/m³, 100,000 µg/m³–101,905 µg/m³, 400,100 µg/m³–410,384 µg/m³, and 100,500 µg/m³–154,977 µg/m³ based on emission data from CAAC, IC, IW, and SP-1 respectively. Maximum biogas concentrations surrounding Aminazar landfill based on four different simulations of LandGEM model were in the sequential order of IW > CAAC > SP-1 > IC which is similar to the findings of Matuail landfill. Maximum biogas concentrations near Aminbazar landfill were higher than Borujerd landfill (Talaiekhosani et al., 2018a) and Shahrekord landfill (Talaiekhosani et al., 2018b) (Table 3). Concentrations of total landfill gas in most areas of Dhaka due to emissions from Aminbazar landfill were estimated to be in the ranges of 10,000 µg/m³–100,000 µg/m³, 8000 µg/m³–80,000 µg/m³, 20,100 µg/m³–200,100 µg/m³, and 8500 µg/m³–80,500 µg/m³ based on emission data from CAAC, IC, IW, and SP-1 respectively.

Due to the complex composition of biogas, a specific maximum ground level concentration has not yet been established. Hence, in order to determine whether this concentration of biogas poses a threat to people, it is necessary to individually model each gas contained in the biogas (Talaiekhosani et al., 2018b).

4.5 Emission of Methane from Matuail and Aminbazar Landfills

Figure 9 displays the estimated emission of methane (CH₄) gas, from both Matuail and Aminbazar landfills. The estimates were generated using four

different simulations of the LandGEM V-3.02 model (CAAC, IC, IW, SP-1).

The model's output indicates that in 2016, the Matuail landfill released 3.81 Gg/year, 1.80 Gg/year, 22.95 Gg/year, and 2.99 Gg/year of CH₄, as anticipated by CAAC, IC, IW, and SP-1 respectively. However, the results suggested that the Aminbazar landfill had lower emission rates in 2016 compared to the Matuail landfill. The CAAC IC, IW, and SP-1 anticipated that the Aminbazar landfill emitted 3.79 Gg/year, 1.79 Gg/year, 22.81 Gg/year, and 2.97 Gg/year of CH₄ in 2016, respectively. Over time, the Aminbazar landfill exhibited higher emission rates than the Matuail landfill, aligning with the findings of Toha and Rahman (2023).

The rate of CH₄ emissions reached its maximum level in 2022 at the Matuail landfill and in 2023 at the Aminbazar landfill. The expected CH₄ emission rates from Matuail landfill in 2022, as estimated by CAAC, IC, IW, and SP-1 are 31.07 Gg/year, 15.05 Gg/year, 73.04 Gg/year, and 23.22 Gg/year respectively. However, according to the CACC, IC, IW, and SP-1 models, the Aminbazar landfill is projected to emit the maximum quantity of CH₄ in 2023. The expected emissions for each model are as follows: CACC—40.29 Gg/year, IC—19.61 Gg/year, IW—78.95 Gg/year, and SP-1—29.81 Gg/year. The LandGEM models anticipated that the peak CH₄ emission rates followed the order of IW > CAAC > SP-1 > IC. Following the

Table 3 Maximum concentrations of different landfill gased emitted from various landfills around the world

Landfills	Country	Total Landfill Gas (Biogas), $\mu\text{g}/\text{m}^3$	CO ₂ , $\mu\text{g}/\text{m}^3$	CH ₄ , $\mu\text{g}/\text{m}^3$	NMOC, $\mu\text{g}/\text{m}^3$	References
Borujerd Landfill, Borujerd	Iran	115,208	82,292	31,380	1317	Talaiekhazani et al. (2018a)
Shahrekord Landfill, Shahrekord	Iran	35,199	25,711	9374	252	Talaiekhazani et al. (2018b)
Sarimukti Landfill, West Java	Indonesia	-	-	21,300	-	Wijaya et al. (2021)
Matuail Landfill, Dhaka	Bangladesh	218,872 (AERMOD-LandGEM-CAAC), 107,038 (AERMOD-LandGEM-IC), 268,604 (AERMOD-LandGEM-IW), 160,304 (AERMOD-LandGEM-SP-1)	160,304 (AERMOD-LandGEM-CAAC), 78,511 (AERMOD-LandGEM-IC), 196,909 (AERMOD-LandGEM-IW), 117,388 (AERMOD-LandGEM-SP-1)	58,568 (AERMOD-LandGEM-CAAC), 28,527 (AERMOD-LandGEM-IC), 71,695 (AERMOD-LandGEM-IW), 42,916 (AERMOD-LandGEM-SP-1)	2524 (AERMOD-LandGEM-CAAC), 1262 (AERMOD-LandGEM-IC), 3029 (AERMOD-LandGEM-IW), 1767 (AERMOD-LandGEM-SP-1)	This study
Aminbazar Landfill, Dhaka	Bangladesh	209,430 (AERMOD-LandGEM-CAAC), 101,905 (AERMOD-LandGEM-IC), 410,384 (AERMOD-LandGEM-IW), 154,977 (AERMOD-LandGEM-SP-1)	153,502 (AERMOD-LandGEM-CAAC), 74,724 (AERMOD-LandGEM-IC), 300,739 (AERMOD-LandGEM-IW), 113,606 (AERMOD-LandGEM-SP-1)	55,928 (AERMOD-LandGEM-CAAC), 27,181 (AERMOD-LandGEM-IC), 109,645 (AERMOD-LandGEM-IW), 41,370 (AERMOD-LandGEM-SP-1)	2396 (AERMOD-LandGEM-CAAC), 1198 (AERMOD-LandGEM-IC), 4699 (AERMOD-LandGEM-IW), 1751 (AERMOD-LandGEM-SP-1)	This study

peak CH₄ emission in 2022 and 2023 for the Matuail and Aminbazar landfills, the emission rate will steadily decline until 2155. The Aminbazar and Matuail landfills were found to have higher peak CH₄ emission rates compared to other landfills such as Borujerd landfill (Talaiekhazani et al., 2018a), Shahrekord landfill (Talaiekhazani et al., 2018b), Akortiri landfill (Chalvatzi & Lazaridis, 2010). Peak CH₄ emission rates in Aminbazar and Matuail landfills estimated using LandGEM-IW model were found to be higher than the emission rates of landfill sites of Delhi-Okhla, Bhalswa and Ghazipur estimated using LandGEM model (Ghosh et al., 2019). The increased CH₄ yield in Matuail and Aminbazar landfills is due to the larger volume of incoming waste and its organic-rich composition (Fig. B.1). CH₄ production arises directly from the anaerobic decomposition of organic substrates; thus, landfills with a higher biodegradable content demonstrate greater CH₄ generation than those with lower organic fractions (Toha & Rahman, 2023).

According to Talaiekhazani et al. (2018a) and Talaiekhazani et al. (2018b) each cubic metre of methane possesses an energy potential of 0.00165 MW. CAAC, IC, IW, and SP-1 model estimated that by 2023, the amount of CH₄ generation from Matuail landfill would be 44,303,650 m³/year, 21,675,038 m³/year, 54,364,363 m³/year, and 32,444,984 m³/year respectively. Nevertheless, achieving a 100% recovery efficiency CH₄ produced by a landfill is unattainable due to limitations in the gas collection equipment. Consequently, a portion of the CH₄ that is generated is consistently released into the atmosphere, while the remainder has the potential to be oxidised in the top cover ((Talaiekhazani et al., 2018a, 2018b). Assuming a CH₄ extraction rate of only 70%, the amount of CH₄ that can be extracted from Matuail landfill is as follows: CAAC—31,012,555 m³/year, IC—15,172,527 m³/year, IW—38,055,054 m³/year, and SP-1—22,711,489 m³/year. Therefore, according to the estimations of CAAC, IC, IW, and SP-1, 51,000 MW, 25,000 MW, 63,000 MW, and 37,500 MW of electrical energy could have been produced in 2023 from Matuail landfill respectively. In case of Aminbazar landfill, CH₄ emissions in 2023 estimated by CAAC, IC, IW, and SP-1 models were 60,384,117 m³/year, 29,392,442 m³/year, 118,337,785 m³/year, and 44,683,778 m³/year respectively, 70% of which are 42,268,882 m³/year, 20,574,709 m³/year, 82,836,450

m³/year, and 31,278,645 m³/year respectively. Hence, estimations of CAAC, IC, IW, and SP-1 revealed that 70,000 MW, 34,000 MW, 137,000 MW, 52,000 MW of electrical energy could have been produced in 2023 from Aminbazar landfill respectively.

The regression analysis of the methane generation constant (K) and peak methane generation for the Matuail landfill resulted in the equation $y = 77.351x + 18.963$, with R² value of 0.9407 (Fig. B.5). This indicates that 94.07% of the variation in the peak value, which is the dependent variable, can be explained by the variation in the methane production constant (K) value. The equation for Aminbazar landfill is $y = 76.055x + 25.812$, with an R² value of 0.9003 (Fig. B.5). This shows that 90.03% of the fluctuation in the peak value (dependent variable) can be explained by the variability in the methane production constant (K) value.

4.6 Dispersion of Methane from Matuail and Aminbazar Landfills

The results of CH₄ dispersion from the Matuail landfill in 2023, obtained using the AERMOD View software, are presented in Fig. 10. These results are based on four simulations of the LandGEM model, namely CAAC, IC, IW, and SP-1. The study findings indicated that the areas surrounding the Matuail landfill have the highest level of CH₄ concentration. Talaiekhazani et al. (2018a), Talaiekhazani et al. (2018b), and Wijaya et al. (2021) have likewise observed the highest levels of CH₄ in the vicinity of dump sites. CAAC, IC, IW, and SP-1 predicted that the CH₄ concentrations surrounding the Matuail landfill varied between 50,000 µg/m³ and 58,568 µg/m³, 20,000 µg/m³ and 28,527 µg/m³, 65,000 µg/m³ and 71,695 µg/m³, and 35,000 µg/m³ and 42,916 µg/m³ respectively. The maximum concentration of CH₄ surrounding the Matuail landfill, as determined by four distinct simulations of the LandGEM model, followed the order of IW > CAAC > SP-1 > IC. The concentrations of CH₄ in most regions of Dhaka, resulting from emissions from the Matuail landfill, were estimated to range from 4000 µg/m³ to 30,000 µg/m³, 1000 µg/m³ to 10,000 µg/m³, 5000 µg/m³ to 50,000 µg/m³, and 1000 µg/m³ to 15,000 µg/m³, based on emission data from CAAC, IC, IW, and SP-1 accordingly. The CH₄ concentrations in the vicinity of the Matuail landfill were determined to be higher than those observed

in other landfills worldwide, including the Borujerd landfill (Talaiekhosani et al., 2018a), the Shahrekord landfill (Talaiekhosani et al., 2018b), and Sarimukti landfill (Wijaya et al., 2021) (Table 3).

Figure 11 displays the outcomes of CH₄ dispersion from the Aminbazar landfill in 2023. These results were obtained using the AERMOD View program and were based on four simulations of the LandGEM model, namely CAAC, IC, IW, and SP-1. The highest levels of CH₄ were found near Aminbazar landfill which ranged from 50,000 µg/m³ to 55,928 µg/m³, 20,000 µg/m³ to 27,181 µg/m³, 100,000 µg/m³ to 109,645 µg/m³, and 35,000 µg/m³ to 41,370 µg/m³ according to emission data from CAAC, IC, IW, and SP-1 respectively. The order of the maximum CH₄ concentrations based on emission data from four different simulations of LandGEM model, from highest to lowest, was IW, CAAC, SP-1, and IC. These findings are consistent with the results obtained from the Matuail landfill. CH₄ concentrations at Aminbazar landfill were higher than those near Borujerd landfill (Talaiekhosani et al., 2018a) and the Shahrekord landfill (Talaiekhosani et al., 2018b), and Sarimukti landfill (Wijaya et al., 2021) (Table 3). The concentrations of CH₄ gas in most areas of Dhaka, resulting from emissions from Aminbazar landfill, were estimated to range from 4000 µg/m³ to 30,000 µg/m³, 1000 µg/m³ to 8000 µg/m³, 80,000 µg/m³ to 1000 µg/m³, and 15,000 µg/m³ to 80,500 µg/m³, based on emission data from CAAC, IC, IW, and SP-1 respectively.

At concentrations of approximately 353.1 g/m³, CH₄ is classified as an asphyxiant gas because it can replace oxygen in the blood (Atia, 2004). The National Institute for Occupational Safety and Health (NIOSH) has established that the permissible exposure limit of CH₄ for workers over an 8-h period is 706,300 µg/m³ (Atia, 2004; Talaiekhosani et al., 2018b). The concentrations of CH₄ in locations near the Matuail and Aminbazar landfills obtained through dispersion modeling based on emission data from four distinct simulations of LandGEM (CAAC, IC, IW, and SP-1) were found to be not exceeding the permissible exposure limit set by NIOSH. Therefore, CH₄ does not pose a health concern for workers in Matuail and Aminbazar landfills.

4.7 Emission of Carbon Dioxide from Matuail and Aminbazar Landfills

Figure 12 displays the estimated emission of carbon dioxide (CO₂), from both Matuail and Aminbazar

landfills. The estimates were generated using four different simulations of the LandGEM V-3.02 model (CAAC, IC, IW, SP-1).

The model's results show that in 2016, the Matuail landfill emitted 10.46 Gg/year of CO₂, as predicted by CAAC. Additionally, the model predicted emissions of 4.94 Gg/year, 62.97 Gg/year, and 8.20 Gg/year of CO₂ from the landfill, as anticipated IC, IW, and SP-1 model, respectively. Nevertheless, the findings indicated that the Aminbazar landfill exhibited lower emission rates in 2016 in contrast to the Matuail landfill. CAAC, IC, IW, and SP-1 predicted that in 2016, the Aminbazar landfill released 10.39 Gg/year, 4.91 Gg/year, 62.58 Gg/year, and 8.14 Gg/year of CO₂, respectively. Over a period of time, the Aminbazar landfill demonstrated elevated rates of emission of CO₂ compared to the Matuail landfill. The CO₂ emission rate peaked in 2022 at the Matuail landfill and in 2023 at the Aminbazar landfill. The projected CO₂ emissions from Matuail landfill in 2022, as calculated by CAAC, IC, IW, and SP-1, are 85.26 Gg/year, 41.30 Gg/year, 200.40 Gg/year, and 63.70 Gg/year, respectively. Based on the CAAC, IC, IW, and SP-1 models, it is predicted that the Aminbazar landfill will release the highest amount of CO₂ in 2023. The anticipated emissions for each model in case of Aminbazar landfill are as follows: The emissions for CAAC are 110.53 Gg/year, for IC are 53.80 Gg/year, for IW are 216.62 Gg/year, and for SP-1 are 81.79 Gg/year. The LandGEM models predicted that the highest rates of CO₂ emissions occurred in the following order: IW > CAAC > SP-1 > IC. After reaching its highest point between 2022 and 2023, the emission of CO₂ from the Matuail and Aminbazar landfills will gradually decrease until 2155. The Aminbazar and Matuail landfills had greater peak CO₂ emission rates in comparison to other landfills, including Borujerd landfill (Talaiekhosani et al., 2018a) and Shahrekord landfill (Talaiekhosani et al., 2018b). The primary factor may be the substantial accumulation of organic waste in Dhaka's landfills, which decomposes efficiently in both aerobic and anaerobic environments (Fig. B.1). The elevated organic matter content facilitates more thorough microbial decomposition, resulting in greater CO₂ emissions compared to landfills with less biodegradable waste composition (Toha & Rahman, 2023). The CO₂ emissions (Fig. 11) estimated from both Aminbazar and Matuail landfills exceed the quantity of CH₄ (Fig. 8) produced in the same year.

The regression analysis of the methane generation constant (K) and peak carbon dioxide generation for the Matuail landfill yielded the equation $y = 212.23x + 52.031$, with R^2 value of 0.9407 (Fig. B.6). This suggests that 94.07% of the variability in the peak value, which is the dependent variable, can be accounted for by the variability in the methane production constant (K) value. The equation representing the Aminbazar landfill is $y = 208.68x + 70.821$. It has R^2 value of 0.9003, as shown in Fig. B.6. This indicates that 90.03% of the variation in the peak value (dependent variable) can be accounted for by the variability in the methane production constant (K) value.

4.8 Dispersion of Carbon Dioxide from Matuail and Aminbazar Landfills

Figure 13 displays the outcomes of CO₂ dispersion from the Matuail landfill in 2023, acquired by using the AERMOD View program. The results are derived from four simulations of the LandGEM model, specifically CAAC, IC, IW, and SP-1. The study results revealed that the regions adjacent to the Matuail landfill exhibit the most elevated concentration of CO₂. Talaiekhazani et al. (2018a) and Talaiekhazani et al. (2018b) have also documented the most elevated concentrations of CO₂ in areas surrounding the landfill sites. According to the predictions made by CAAC, IC, IW, and SP-1, the concentrations of CO₂ around the Matuail landfill ranged from 10,600 µg/m³ to 160,304 µg/m³, 70,000 µg/m³ to 78,511 µg/m³, 100,900 µg/m³ to 196,909 µg/m³, and 100,100 µg/m³ to 117,388 µg/m³ correspondingly. The concentrations of CO₂ estimated by four different simulations of LandGEM model followed the order of IW > CAAC > SP-1 > IC. The concentrations of CO₂ in most areas of Dhaka, caused by emissions from the Matuail landfill, were estimated to vary from 8600 µg/m³ to 80,600 µg/m³, 5000 µg/m³ to 50,000 µg/m³, 8900 µg/m³ to 80,900 µg/m³, and 8100 µg/m³ to 80,100 µg/m³, based on emission data from CAAC, IC, IW, and SP-1 respectively. The CO₂ concentrations near the Matuail landfill were found to be higher than those found in other landfills around the world, such as Borujerd landfill (Talaiekhazani et al., 2018a) and the Shahrekord landfill (Talaiekhazani et al., 2018b) (Table 3).

Figure 14 illustrates the results of the dispersion of CO₂ from the Aminbazar landfill in the year 2023. The data were acquired through using AERMOD View

software and were derived from four simulations of the LandGEM model, specifically CAAC, IC, IW, and SP-1. The maximum concentrations of CO₂ were found in the areas surrounding the Aminbazar landfill, ranging from 100,500 µg/m³ to 153,502 µg/m³, 65,000 µg/m³ to 74,724 µg/m³, 300,000 µg/m³ to 300,739 µg/m³, and 100,100 µg/m³ to 113,607 µg/m³, as reported by CAAC, IC, IW, and SP-1 respectively. The maximum CO₂ concentrations, based on emission data from four distinct simulations of the LandGEM model, were ranked in the following order from highest to lowest: IW > CAAC > SP-1 > IC. These findings align with the outcomes derived from the Matuail landfill. The concentrations of CO₂ in Aminbazar landfill were more than those at Borujerd landfill (Talaiekhazani et al., 2018a) and the Shahrekord landfill (Talaiekhazani et al., 2018b) (Table 3). The levels of CO₂ in most parts of Dhaka, caused by emissions from the Aminbazar landfill, were estimated to vary between 8500 µg/m³ and 80,500 µg/m³, 5000 µg/m³ and 50,000 µg/m³, 10,000 µg/m³ and 100,000 µg/m³, and 8100 µg/m³ and 80,100 µg/m³, based on emission data from CAAC, IC, IW, and SP-1 respectively.

CO₂, a greenhouse gas, is commonly released into the atmosphere from sources such as municipal solid waste landfills. This study pertains to CO₂ emissions originating from the Matuail and Aminbazar landfills. It does not take into account other sources of CO₂, such as transportation as well as industrial sources. Carbon dioxide can cause irritation to the eyes, nose, and throat when present in high amounts (OSHA, 2017). According to NIOSH, the recommended exposure limit for carbon dioxide is 9,000,000 µg/m³ (OSHA, 2017). The concentrations of CO₂ in areas adjacent to the Matuail and Aminbazar landfills, as determined by dispersion modelling using emission data from four different LandGEM simulations (CAAC, IC, IW, and SP-1), were found to be within the allowable exposure level established by NIOSH. Thus, CO₂ emitted from the landfill does not cause any health risks to the workers and residents living near the landfill sites.

4.9 Emission of NMOCs from Matuail and Aminbazar Landfills

Figure 15 illustrates the estimated non-methane organic carbons (NMOCs) emissions from the Matuail and Aminbazar landfills. The estimations were derived from four distinct simulations conducted

Table 4 Concentrations of different air pollutants measured from different sampling points near Matuail and Aminbazar landfills using Haz-Scanner-TM model HIM-600

Location	Sampling Points	GPS Coordinates	CH ₄ (ppm)	CO ₂ (ppm)	CO (ppb)	NO (ppb)	NO ₂ (ppb)	SO ₂ (ppb)	O ₃ (ppb)	PM1 (µg/m ³)	PM2.5 (µg/m ³)
Matuail	S1	23°42'59.6"N, 90°27'0.5"E	161.25	526.28	177.3	13.39	15.91	15.89	39.31	58.24	80.25
Matuail	S2	23°42'59.5"N, 90°27'10.4"E	195.5	525.93	140.99	61.79	2	22.95	2.1	6.67	10.82
Matuail	S3	23°42'49.5"N, 90°27'3.3"E	145.9	552.02	779.44	66.17	2	30.37	4.85	72.31	76.25
Average			167.55	534.74	365.91	47.11	6.64	23.07	15.42	45.74	55.77
Aminbazar	S4	23°47'42.83"N, 90°18'1.54"E	278.4	682.2	213.18	18.72	63.7	23.02	62.66	57.42	86.2
Aminbazar	S5	23°47'39.86"N, 90°17'47.17"E	162.9	530.01	1293.74	124.85	2	40.5	5.98	18.83	35.03
Aminbazar	S6	23°47'34.13"N, 90°18'7.01"E	105	566.9	251.13	106.9	2	50.3	1	6.54	25.7
Average			182.1	593.04	586.02	83.49	22.57	37.94	23.21	27.59	48.98
WHO Guideline (WHO, 2005)			-	-	10,000 (8 h)	-	21.27 (annual)	7.64 (24 h)	50.97 (8 h)	-	10 (annual)
Bangladesh Standard (ACPR, 2022)			-	-	9000 (8 h)	53 (annual)	53 (annual)	30 (annual)	80 (8 h)	-	15 (annual)

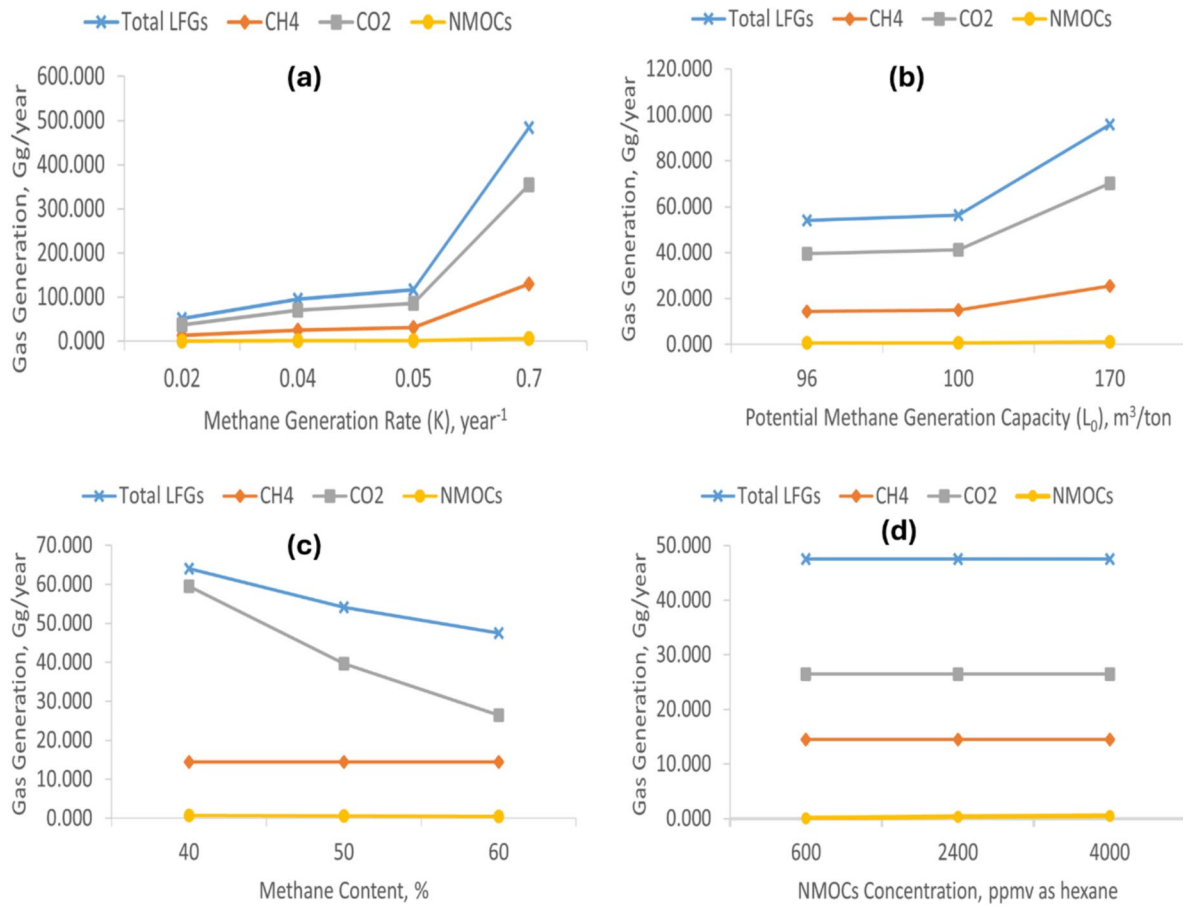


Fig. 6 Uncertainty analysis for (a) methane generation rate (K), (b) potential methane generation capacity (L₀), (c) methane content, (d) potential NMOCs concentration for Matuail landfill

using the LandGEM V-3.02 model (CAAC, IC, IW, SP-1).

According to the model's findings, the Matuail landfill released 0.16 Gg/year of NMOCs in 2016, as projected by the CAAC. In addition, the model forecasted emissions of 0.08 Gg/year, 0.99 Gg/year, and 0.13 Gg/year of NMOCs from the landfill, as predicted by the IC, IW, and SP-1 models, respectively. The Aminbazar landfill emitted similar amount of NMOCs in 2016 as predicted by different simulations of LandGEM model except IW (0.98 Gg/year). Over a duration, the Aminbazar landfill exhibited higher levels of NMOCs emission in comparison to the Matuail landfill. The rate of NMOCs emissions reached its highest point in 2022 at the Matuail landfill and in 2023 at the

Aminbazar landfill. The estimated NMOCs emissions from Matuail landfill in 2022, as determined by CAAC, IC, IW, and SP-1, are 1.34 Gg/year, 0.65 Gg/year, 3.14 Gg/year, and 1.00 Gg/year, respectively. According to the CACC, IC, IW, and SP-1 models, it is projected that the Aminbazar landfill will emit the most quantity of NMOCs in the year 2023. The projected emissions for each model in the event of Aminbazar landfill are as follows: The emissions of CACC amount to 1.73 Gg/year, whereas IC predicted 0.84 Gg/year. IW predicted emissions of 3.39 Gg/year, and SP-1 predicted 1.28 Gg/year. The emission levels of NMOCs predicted by IW is greater than CAAC, which is greater than SP-1, which is greater than IC. Following its peak between 2022 and 2023, the NMOCs emissions

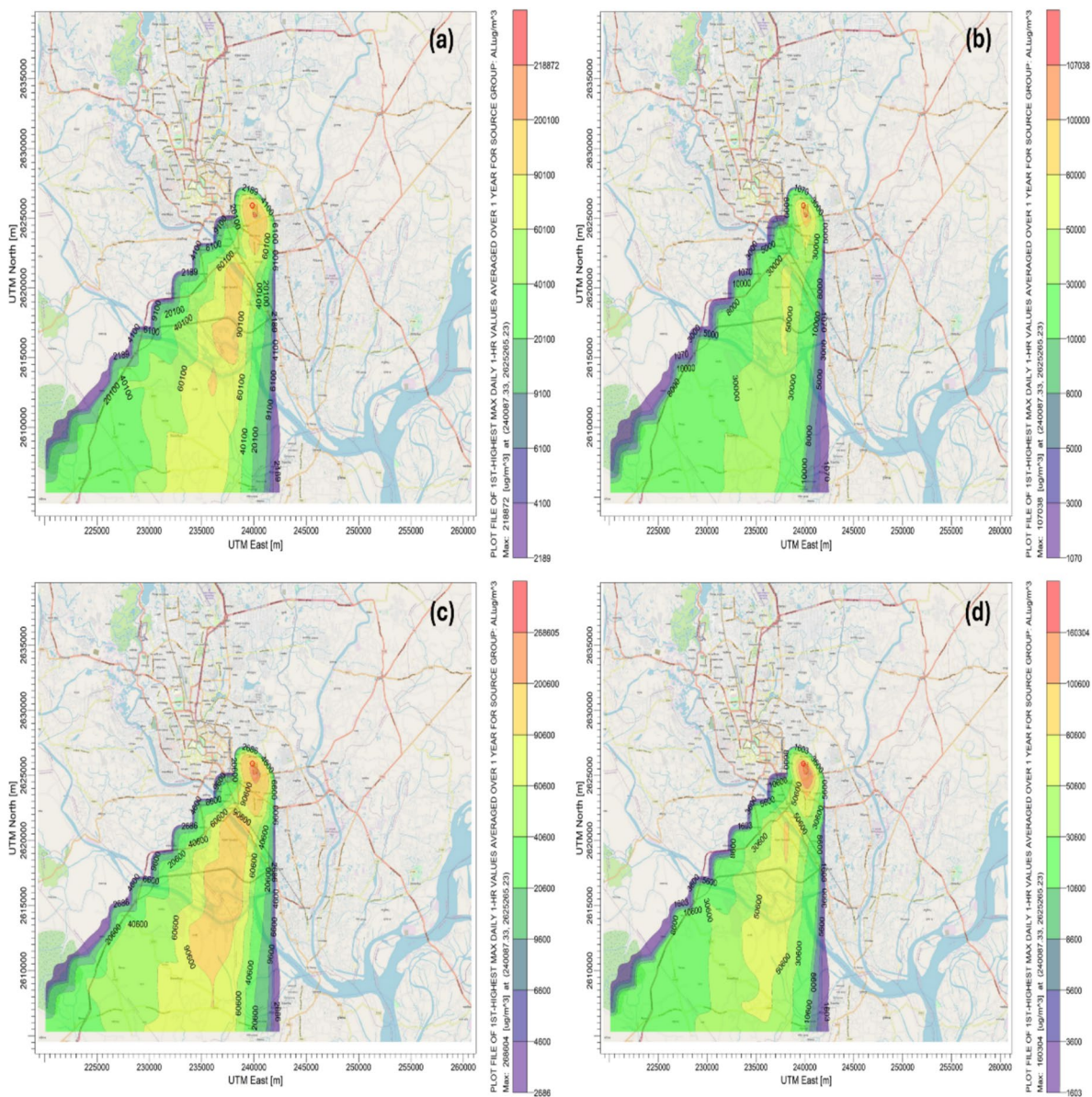


Fig. 7 Results of biogas dispersion from Matuail landfill in 2023 using AERMOD View software based on emission data from (a) CAAC, (b) IC, (c) IW, and (d) SP-1 of LandGEM V-3.02 model

from the Matuail and Aminbazar landfills will progressively decline until 2155. The Aminbazar and Matuail landfills exhibited higher peak NMOCs emission rates compared to other landfills, such as Borujerd landfill (Talaiekhosani et al., 2018a) and Shahrekord landfill (Talaiekhosani et al., 2018b). This difference is likely associated with the higher proportion of decomposable waste streams (Toha & Rahman, 2023) that release volatile organic

compounds during decomposition. The persistent and elevated rate of waste disposal maintains conditions conducive to NMOC release, which accounts for the heightened emissions observed in Dhaka's landfills (Fig. B.1).

While the total quantity of NMOCs is lower than the amounts of CO_2 and CH_4 , many NMOCs are extremely poisonous and pose a significant risk to human health (Omrani et al., 2009). Table A.5

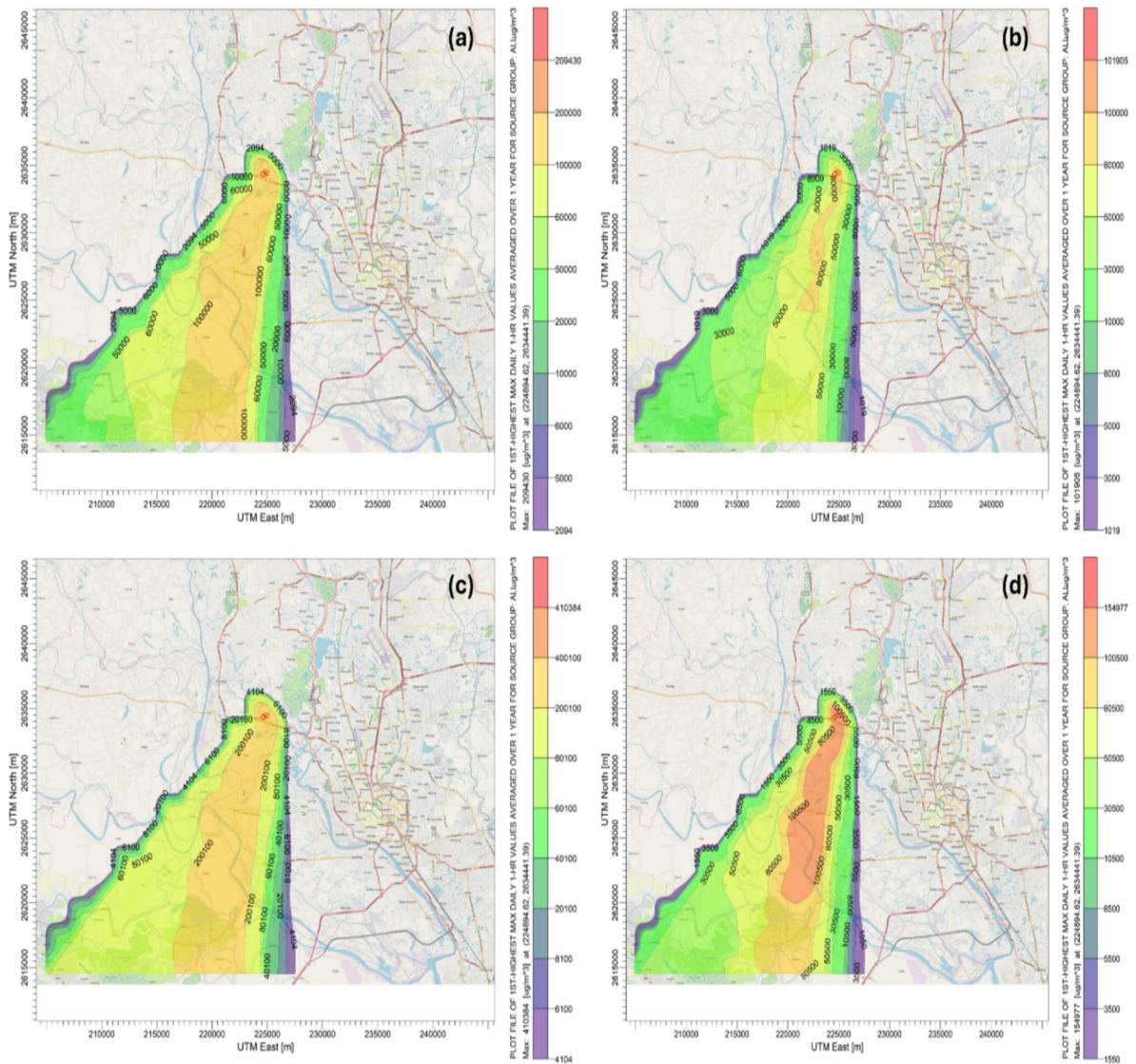


Fig. 8 Results of biogas dispersion from Aminbazar landfill in 2023 using AERMOD View software based on emission data from (a) CAAC, (b) IC, (c) IW, and (d) SP-1 of LandGEM V-3.02 model

displays a comprehensive list of 48 non-methane organic compounds (NMOCs) that have been predicted to be generated from the Matuail landfill in the year 2023 and Table A.6 shows the list of 48 non-methane organic compounds (NMOCs) that have been predicted to be generated from the Aminbazar landfill in the year 2023. Based on the predictions made by CAAC, IC, IW, and SP-1, in 2023, the emission rates of benzene (a carcinogenic substance) from Matuail landfill were 0.0037135 Gg/year, 0.0018168 Gg/year, 0.0045568 Gg/year, and 0.0027195 Gg/

year respectively and In case of Aminbazar landfill, the emission rates of benzene in 2023 were comparatively higher as compared of Matuail landfill. The emission rates of benzene from Aminbazar landfill in 2023 were 0.0050614 Gg/year, 0.0024637 Gg/year, 0.0099190 Gg/year, and 0.0037454 Gg/year, as predicted by CAAC, IC, IW, and SP-1 respectively.

The regression analysis of the methane generation constant (K) and peak NMOc generation for the Matuail landfill resulted in the equation $y = 3.3247x + 0.8151$, with R^2 value of 0.9407

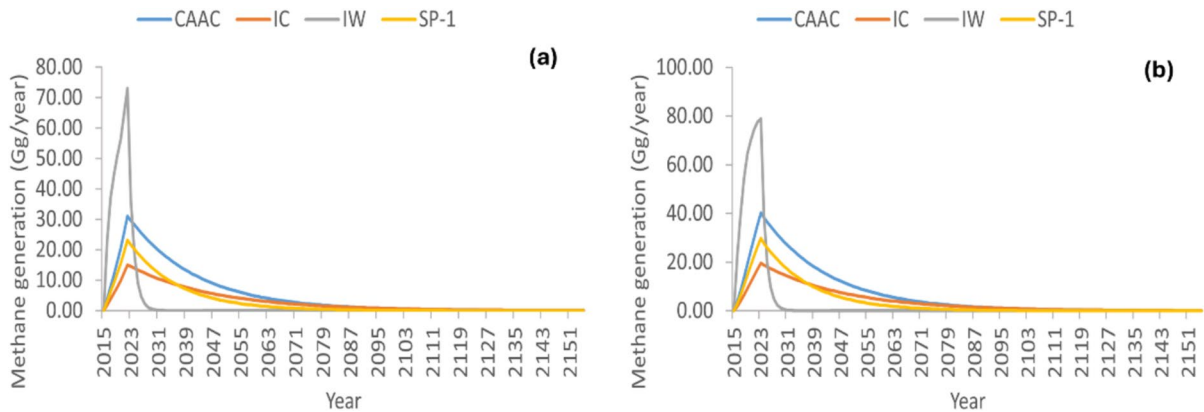


Fig. 9 Graph showing methane gas generation rate predicted by four simulations of LandGEM V-3.02 model from (a) Matuail landfill and (b) Aminbazar landfill

(Fig. B.7). This indicates that 94.07% of the variation in the peak value, which is the dependent variable, can be explained by the variation in the methane generation constant (K) value. Regression analysis in case Aminbazar landfill can be mathematically represented by the equation $y = 3.2691x + 1.1094$. The R^2 value is 0.9003, as depicted in Fig. B.7. This suggests that 90.03% of the fluctuation in the peak value, which is the dependent variable, can be explained by the change in the methane generation constant (K) value.

4.10 Dispersion of NMOCs from Matuail and Aminbazar Landfills

Figure 16 exhibits the results of non-methane organic compounds (NMOCs) dispersion from the Matuail landfill in 2023, obtained by the utilization of the AERMOD View software. The results are obtained from four simulations of the LandGEM model, namely CAAC, IC, IW, and SP-1. The study findings indicated that the areas neighboring the Matuail landfill have the highest levels of NMOCs concentration. Talaiekhazani et al. (2018a) and Talaiekhazani et al. (2018b) have also reported the maximum levels of NMOC in the vicinity of Borujerd and Shahrekord landfill sites respectively. The CAAC, IC, IW, and SP-1 predicted that the amounts of NMOCs near the Matuail landfill varied between $2500 \mu\text{g}/\text{m}^3$ and $2524 \mu\text{g}/\text{m}^3$, $1000 \mu\text{g}/\text{m}^3$ and $1252 \mu\text{g}/\text{m}^3$, $3000 \mu\text{g}/\text{m}^3$ and $3029 \mu\text{g}/\text{m}^3$, and $1500 \mu\text{g}/\text{m}^3$ and $1767 \mu\text{g}/\text{m}^3$ respectively. The concentrations of NMOCs, as predicted by four simulations

of the LandGEM model, exhibited the following order: $\text{IW} > \text{CAAC} > \text{SP-1} > \text{IC}$. The NMOCs concentrations in most areas of Dhaka, resulting from emissions from the Matuail landfill, were estimated to range from $90 \mu\text{g}/\text{m}^3$ to $1000 \mu\text{g}/\text{m}^3$, $80 \mu\text{g}/\text{m}^3$ to $800 \mu\text{g}/\text{m}^3$, $100 \mu\text{g}/\text{m}^3$ to $1500 \mu\text{g}/\text{m}^3$, and $80 \mu\text{g}/\text{m}^3$ to $1000 \mu\text{g}/\text{m}^3$, based on emission data from CAAC, IC, IW, and SP-1 respectively. The levels of NMOCs around the Matuail landfill were discovered to be greater than those observed in other landfills worldwide, such as the Borujerd landfill (Talaiekhazani et al., 2018a) and the Shahrekord landfill (Talaiekhazani et al., 2018b) (Table 3).

Figure 17 depicts the outcomes of the dispersion of non-methane organic compounds (NMOCs) originating from the Aminbazar landfill in the year 2023. The data were obtained by utilizing the AERMOD View software and were taken from four simulations of the LandGEM model, namely CAAC, IC, IW, and SP-1. Like Matuail landfill, the highest levels of NMOCs were detected in the vicinity of the Aminbazar landfill. The values ranged from $2000 \mu\text{g}/\text{m}^3$ to $2396 \mu\text{g}/\text{m}^3$, $1000 \mu\text{g}/\text{m}^3$ to $1198 \mu\text{g}/\text{m}^3$, $4500 \mu\text{g}/\text{m}^3$ to $4699 \mu\text{g}/\text{m}^3$, and $1500 \mu\text{g}/\text{m}^3$ to $1751 \mu\text{g}/\text{m}^3$, according to the predictions of CAAC, IC, IW, and SP-1 respectively. The NMOCs concentration predicted by IW is greater than CAAC, which is greater than SP-1, which is greater than IC. These findings are consistent with the results obtained from the Matuail landfill. The NMOCs concentrations in Aminbazar landfill exceeded those found in Borujerd landfill (Talaiekhazani et al., 2018a) and the Shahrekord

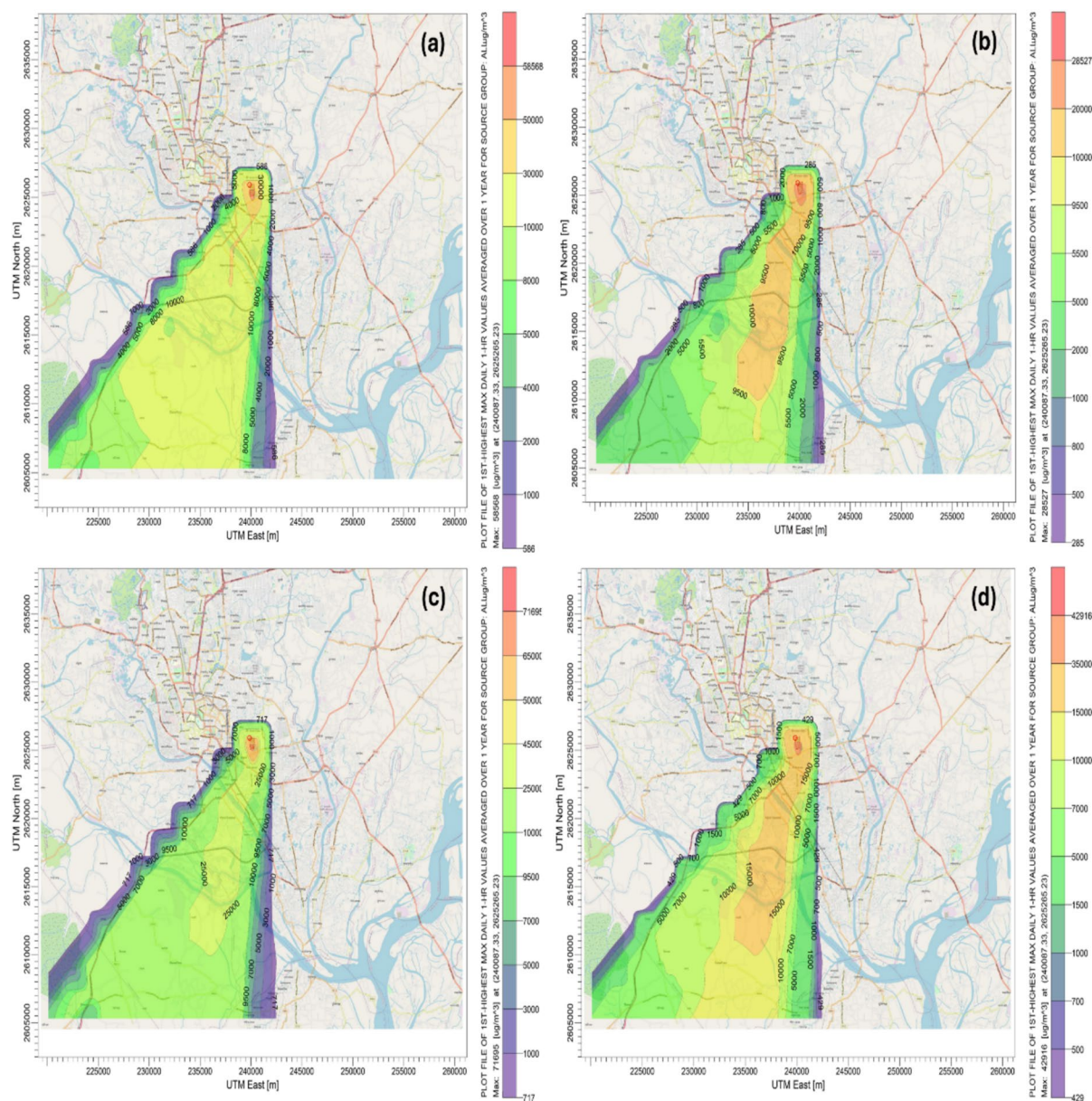
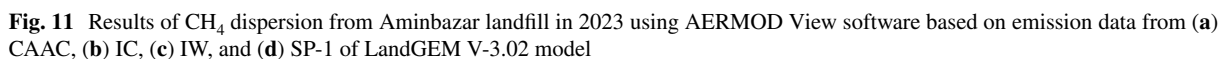


Fig. 10 Results of CH₄ dispersion from Matuail landfill in 2023 using AERMOD View software based on emission data from (a) CAAC, (b) IC, (c) IW, and (d) SP-1 of LandGEM V-3.02 model

landfill (Talaiekhosani et al., 2018b) (Table 3). The levels of NMOCs in most areas of Dhaka, resulting from emissions from the Aminbazar landfill, were estimated to range from 90 µg/m³ to 1000 µg/m³, 80 µg/m³ to 800 µg/m³, 100 µg/m³ to 1000 µg/m³, and 80 µg/m³ to 1000 µg/m³, based on emission data from CAAC, IC, IW, and SP-1 respectively.

Non-methane organic compounds (NMOCs) encompass a wide range of chemical substances, consisting of several compounds. Among these, benzene and vinyl chloride are particularly hazardous. While the levels of non-methane organic compounds (NMOCs) are generally low in Dhaka, some molecules like benzene and vinyl chloride can pose



sites are shown in Fig. B.9 and Fig. B.10. respectively, which indicated that maximum concentrations of benzene were detected near both the landfills and concentrations decreased with distance. Based on the maximum concentrations of benzene near Matuail and Aminbazar landfill sites, health risk assessment was carried out (Liu et al., 2016; USEPA, 2011a, b). The

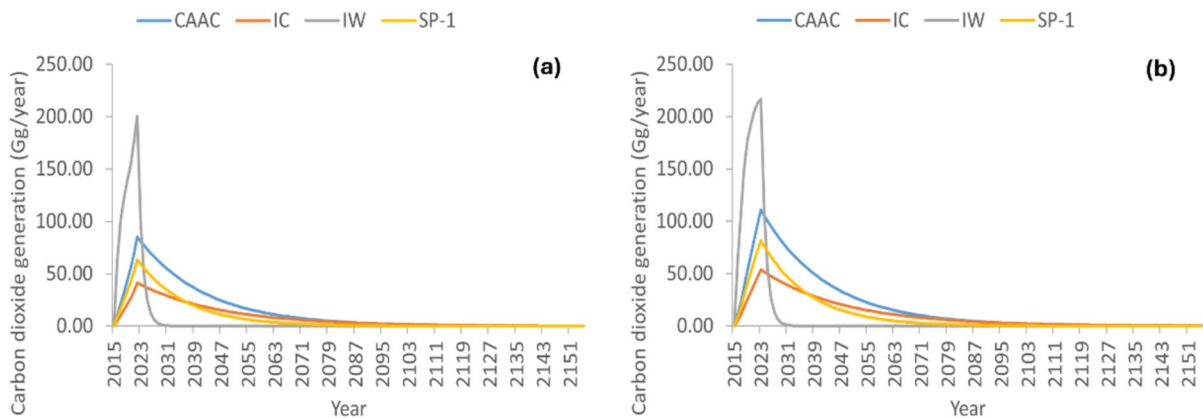


Fig. 12 Graph showing carbon dioxide generation rate predicted by four simulations of LandGEM V-3.02 model from (a) Matuail landfill and (b) Aminbazar landfill

method used for health risk assessment has been provided in the supplementary material. Results of health risk assessment (Table A.7) revealed that inhalation of benzene emitted from Matuail and Aminbazar landfills might pose unacceptable non-carcinogenic as well as carcinogenic risks to workers and nearby residents.

4.11 Field Measurement Findings

Table 4. shows the concentrations of different air pollutants such as methane (CH_4), carbon dioxide (CO_2), carbon monoxide (CO), nitric oxide (NO), nitrogen dioxide (NO_2), sulfur dioxide (SO_2), ozone (O_3), PM_{10} , and $\text{PM}_{2.5}$ measured from different sampling points (Fig. B.2, Fig. B.3) near Matuail and Aminbazar landfills using Haz-Scanner-TM model HIM-600.

Concentrations of methane (CH_4) measured using Haz-Scanner-TM model HIM-600 in different sampling points near Matuail and Aminbazar landfills ranged from 145.9 ppm to 195.5 ppm and 105 ppm to 278.4 ppm respectively, with average concentrations of 167.55 ppm and 182.1 ppm near Matuail and Aminbazar landfills respectively. Anaerobic decomposition of organic wastes could be one of the major anthropogenic sources of methane (CH_4) in the atmosphere near Matuail and Aminbazar landfill sites (Heilig, 1994). The average concentrations of CH_4 near Matuail and Aminbazar landfill sites higher than the concentrations found near landfill sites located in

Taiwan (Chen et al., 2008) and in Lahore (Raza et al., 2021).

Concentration of carbon dioxide (CO_2) near Matuail and Aminbazar landfill sites ranged from 525.93 ppm to 552.02 ppm and 530.01 ppm 682.2 ppm. Average concentrations of CO_2 were found to be higher in case of Aminbazar landfill (593.04 ppm) as compared to Matuail landfill (534.74 ppm). Average concentrations of CO_2 found near both Matuail and Aminbazar landfills exceeded the concentrations found by Chen et al. (2008) and Raza et al. (2021) near waste disposal sites located in Taiwan and Lahore respectively. The possible anthropogenic sources of CO_2 in the atmosphere near Matuail and Aminbazar landfill sites could be decomposition of organic waste in the landfills (Talaiekhosani et al., 2018b) as well as fossil fuel combustion in transportation as well as industrial sectors (Thurston, 2008).

Average concentrations of carbon monoxide (CO) near Matuail and Aminbazar landfills were found to be 365.91 ppb and 586.02 ppb respectively, with co concentrations of CO ranging from 177.3 ppb to 779.44 ppb and 213.18 ppb to 1293.74 ppb near Matuail and Aminbazar landfills respectively. Average concentrations of CO near both Matuail and Aminbazar landfills were within the 8-h period standard of (WHO, 2005). Average CO concentrations near both Matuail and Aminbazar landfills were also within the 8-h period Bangladesh standard established by (APCR, 2022). The average concentrations of CO near Matuail and Aminbazar landfills were found to

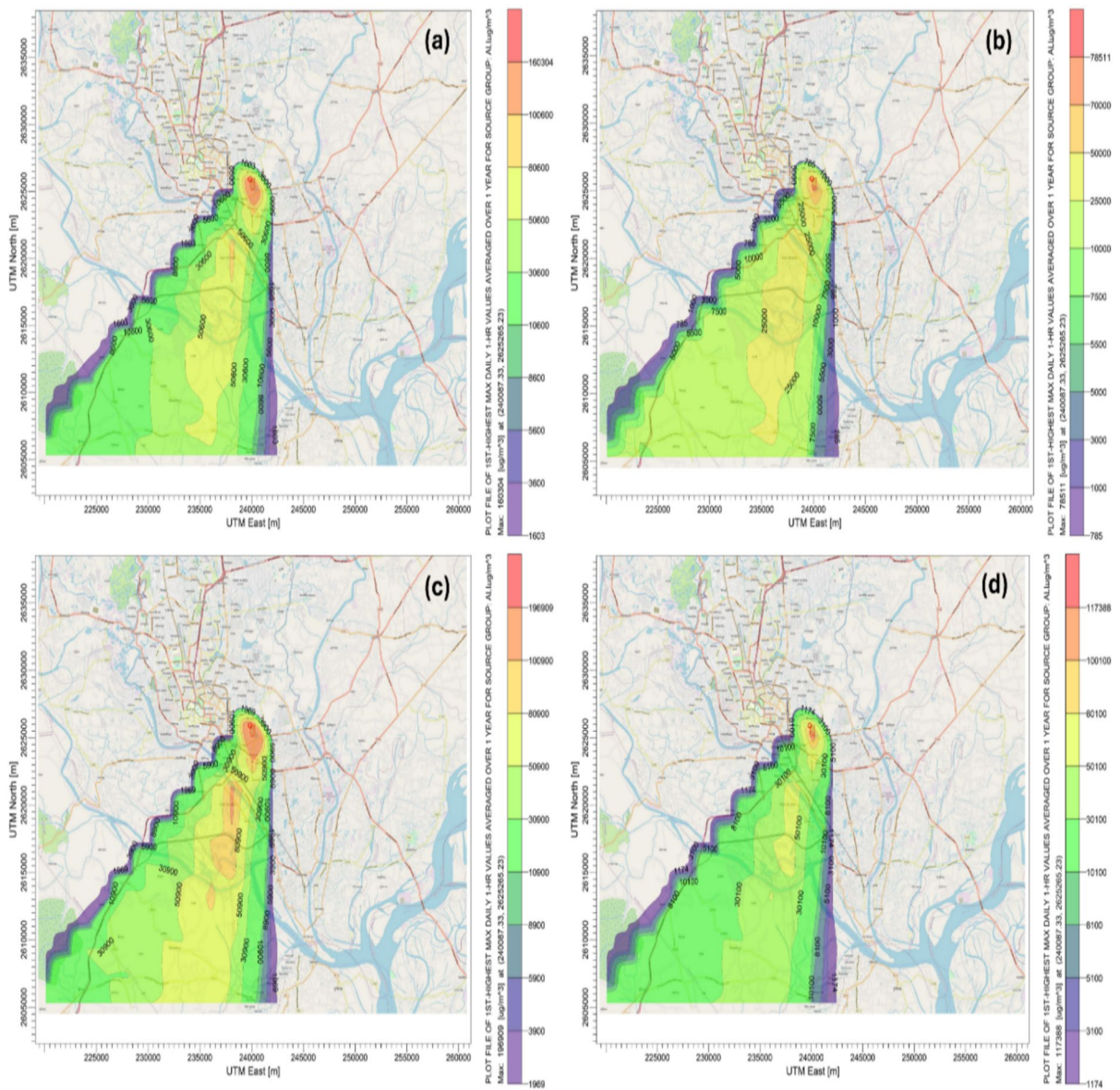


Fig. 13 Results of CO₂ dispersion from Matuail landfill in 2023 using AERMOD View software based on emission data from (a) CAAC, (b) IC, (c) IW, and (d) SP-1 of LandGEM V-3.02 model

be lower than the findings of (Hossain et al., 2019). Non-transportation fuel combustion, industrial processes, and burning of wastes in the landfills could be possible anthropogenic sources of CO (Miller, 2011). In addition, landfill gases emitted into the atmosphere due to decomposition of organic wastes can be converted by ultraviolet radiation into gases like carbon monoxide (CO) (Kumar et al., 2004).

Concentrations of nitric oxides (NO) near Matuail and Aminbazar landfills were found to be in the range of 13.39 ppb–66.17 ppb and 18.72 ppb–124.85 ppb with average concentrations of 47.11 ppb and 83.49 ppb near Matuail and Aminbazar landfills respectively and concentrations of nitrogen dioxide (NO₂) near Matuail and Aminbazar landfills were found to be in the range of 2 ppb–15.91 ppb

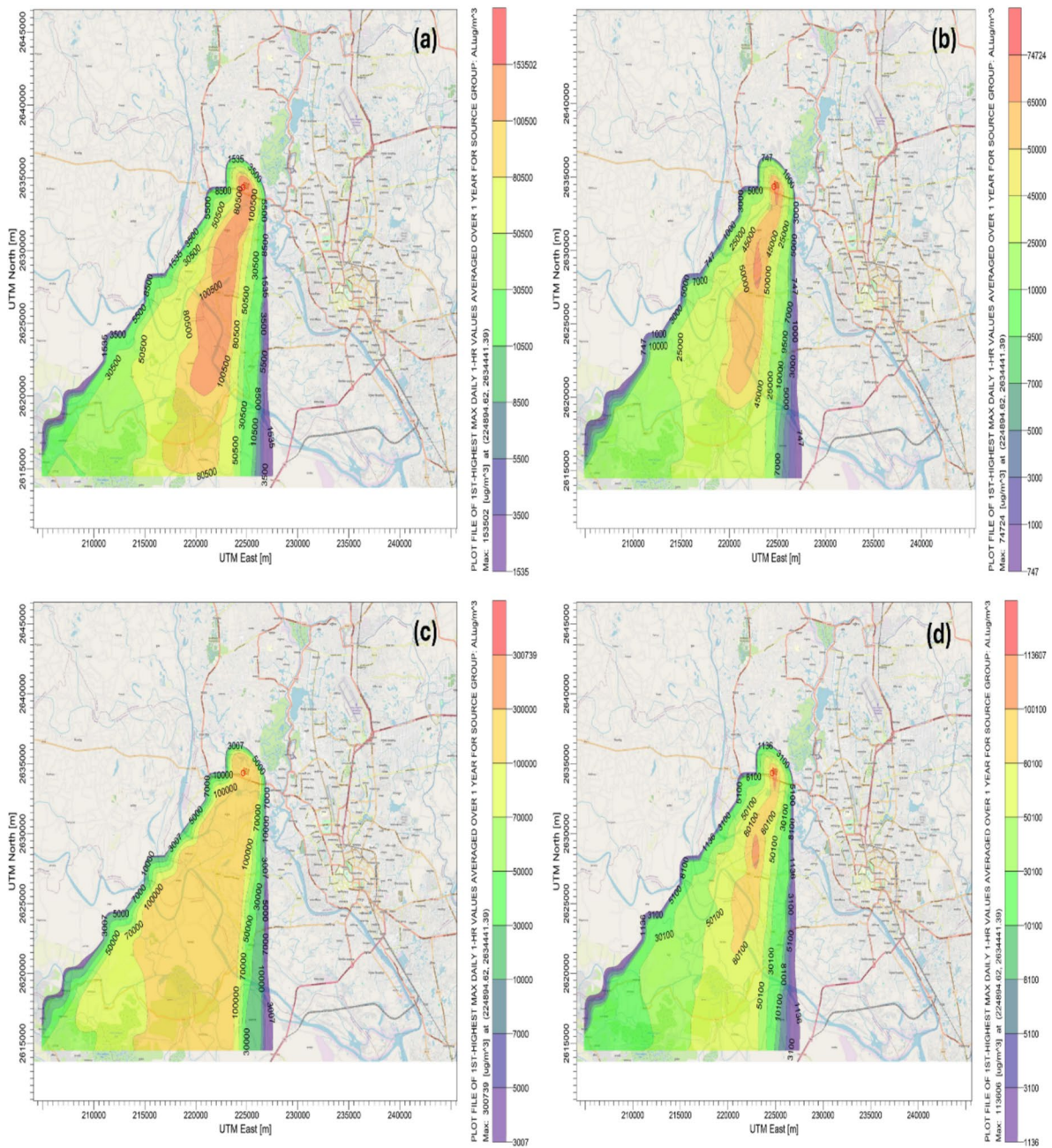


Fig. 14 Results of CO₂ dispersion from Aminbazar landfill in 2023 using AERMOD View software based on emission data from (a) CAAC, (b) IC, (c) IW, and (d) SP-1 of LandGEM V-3.02 model

and 2 ppb—63.7 ppb with average concentrations of 6.64 ppb and 22.57 ppb near Matuail and Aminbazar landfills respectively. Average concentration of NO₂ near Matuail landfill were lower than the annual standard set for NO_x by WHO (2005) and APCR

(2022). Average concentration of NO₂ near Aminbazar landfill exceeded the annual standard of WHO (2005) but was lower than the Bangladesh standard set by APCR (2022). Combustion of fossil fuels in automobiles as well as biomass burning could be

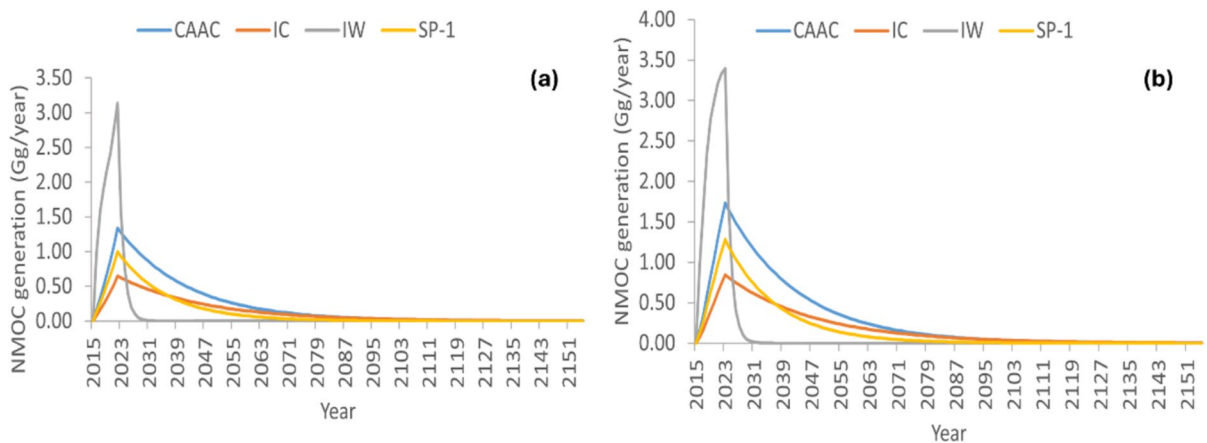


Fig. 15 Graph showing non-methane organic carbons generation rate predicted by four simulations of LandGEM V-3.02 model from (a) Matuail landfill and (b) Aminbazar landfill

the possible anthropogenic sources of NO_x near Matuail and Aminbazar landfill sites (Ghude et al., 2014; Vinken et al., 2014; Wang et al., 2021).

Average concentrations of sulfur dioxide (SO_2) near Matuail and Aminbazar landfills were measured to be 23.07 ppb and 37.94 ppb respectively, with concentrations ranging from 15.89 ppb–30.37 ppb and 23.02 ppb–50.3 ppb adjacent to Matuail and Aminbazar landfills respectively. Average concentration of SO_2 near Matuail landfill exceeded the 24-h period standard of WHO (2005) but was lower than the annual Bangladesh standard set by APCR (2022). On the other hand, average concentration of SO_2 near Aminbazar landfill exceeded both the 24-h period standard of WHO (2005) and annual standard of APCR (2022). Industrial activities burning fossil fuels could be a possible anthropogenic source of SO_2 in the atmosphere (Jyethi, 2016; Vestreng et al., 2007).

The ranges of ozone (O_3) concentrations near Matuail and Aminbazar landfills were 2.1 ppb–39.31 ppb and 1 ppb–62.66 ppb respectively, with mean concentrations of 15.42 ppb and 23.21 ppb near Matuail and Aminbazar landfills respectively. Average concentrations of O_3 near both Matuail and Aminbazar landfills were within the 8-h period standards set by WHO (2005) and APCR (2022). According to Wijaya et al. (2021) if methane (CH_4) emitted from the anaerobic decomposition of organic wastes in landfills is released straight into the atmosphere without any additional control measures, it can

potentially create ozone (O_3). Ozone (O_3) is a type of secondary air pollutant, and the ground level production of ozone (O_3) causes significant environmental problems due to its high toxicity.

Concentrations of PM_{10} near Matuail and Aminbazar landfills were found to be ranging from 6.67 $\mu\text{g}/\text{m}^3$ to 72.31 $\mu\text{g}/\text{m}^3$ and 6.54 $\mu\text{g}/\text{m}^3$ –47.42 $\mu\text{g}/\text{m}^3$ respectively while concentrations of $\text{PM}_{2.5}$ near Matuail and Aminbazar landfills were found to be ranging from 10.82 $\mu\text{g}/\text{m}^3$ to 80.25 $\mu\text{g}/\text{m}^3$ and 25.7 $\mu\text{g}/\text{m}^3$ to 86.2 $\mu\text{g}/\text{m}^3$ respectively. Mean concentrations of PM_{10} and $\text{PM}_{2.5}$ near Matuail landfill were 45.74 $\mu\text{g}/\text{m}^3$ and 55.77 $\mu\text{g}/\text{m}^3$ respectively while mean concentrations of PM_{10} and $\text{PM}_{2.5}$ near Aminbazar landfill were 27.59 $\mu\text{g}/\text{m}^3$ and 48.98 $\mu\text{g}/\text{m}^3$ respectively. The average concentrations of $\text{PM}_{2.5}$ near both Matuail and Aminbazar landfills were measured to exceed the both the annual standards set by WHO (2005) and APCR (2022). A previous study conducted by Hosain et al. (2019) on Matuail landfill site found lower levels of $\text{PM}_{2.5}$ as compared to the present study.

4.12 Comparison between Field Measurements and Model Outputs

The concentrations of methane (CH_4) and carbon dioxide (CO_2) surrounding both Matuail and Aminbazar landfills obtained using AERMOD View air pollution dispersion modeling based on emission date from four distinct simulations of Landgem V-3.02 model was compared with the measured

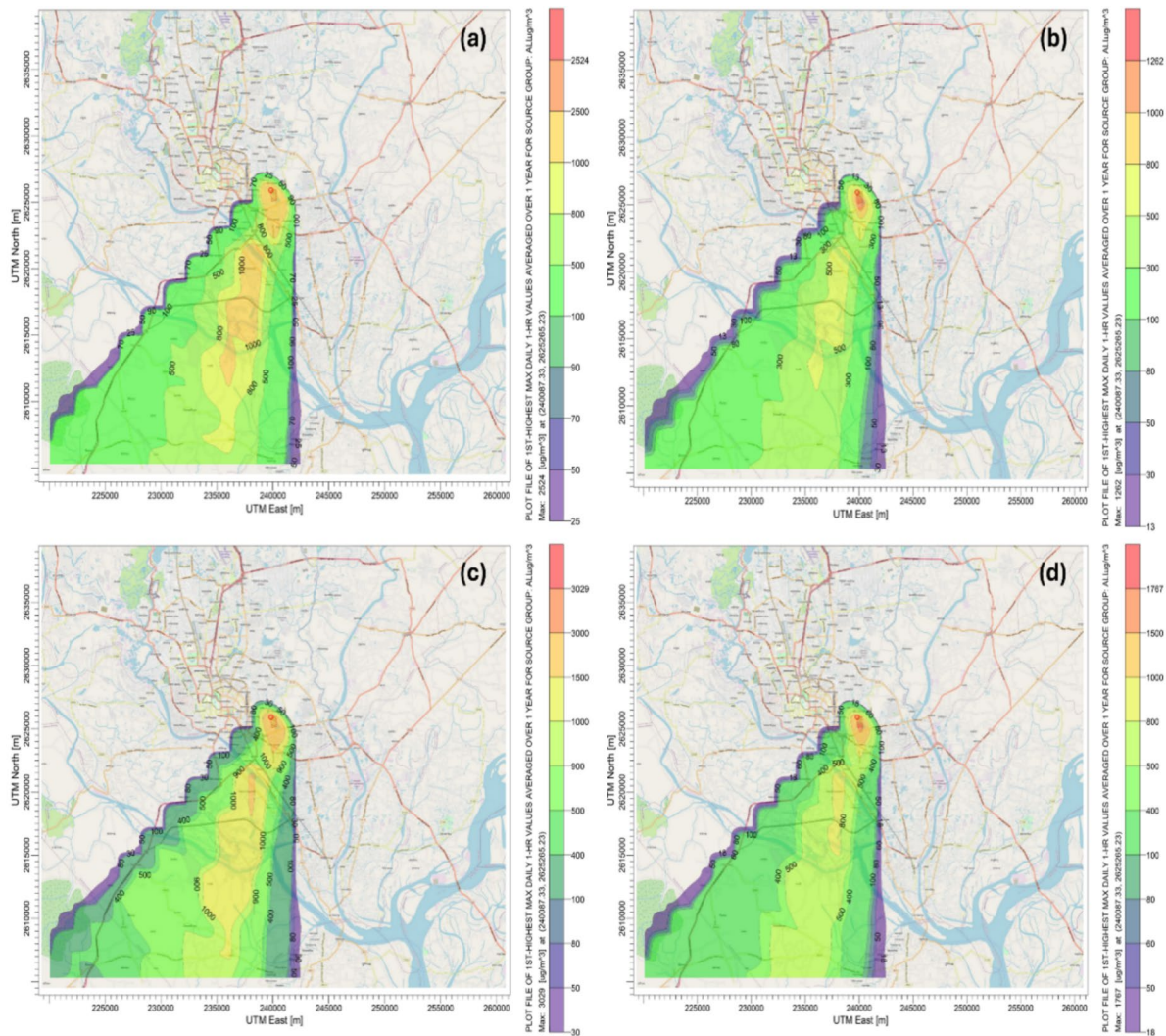


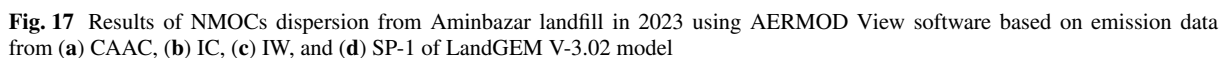
Fig. 16 Results of NMOCs dispersion from Matuail landfill in 2023 using AERMOD View software based on emission data from (a) CAAC, (b) IC, (c) IW, and (d) SP-1 of LandGEM V-3.02 model

concentrations of methane (CH_4) and carbon dioxide (CO_2) using Haz-Scanner-TM model HIM-600 (Fig. 18).

The average concentrations of CH_4 and CO_2 surrounding Matuail landfill obtained by field measurements were 167.55 ppm and 534.74 ppm respectively (Table 3). As compared to concentrations of CH_4 obtained by field measurements, AERMOD View dispersion modeling based on four simulations of LandGEM V-3.02 i.e. CAAC, IC, IW, and SP-1 underestimated the concentrations by 46.71%, 74.05%, 34.77% and 60.95% respectively for Matuail

landfill. In case of the concentrations of CO_2 surrounding Matuail landfill, AERMOD View dispersion modeling based on four simulations of LandGEM V-3.02 i.e. CAAC, IC, IW, and SP-1 underestimated the concentrations by 83.35%, 91.84%, 79.54% and 87.80% respectively as compared to concentrations of CO_2 obtained by field measurements.

Field measurements also revealed that the average concentrations of CH_4 and CO_2 surrounding Aminbazar landfill obtained by field measurements were 182.1 ppm and 593.04 ppm respectively (Table 3). In comparison to concentrations of CH_4



LandGEM V-3.02 i.e. CAAC, IC, IW, and SP-1 underestimated the concentrations by 88.10%, 93.46%, 71.86% and 89.99% respectively in comparison to concentrations of CO₂ obtained by field measurements.

Comparative analysis between the concentrations of methane (CH_4) and carbon dioxide (CO_2) surrounding Matuail and Aminbazar landfills obtained

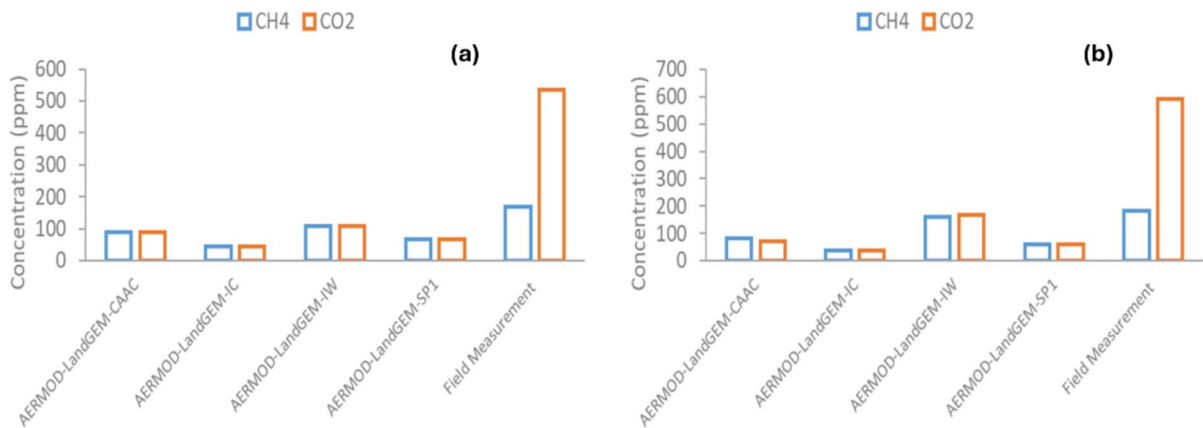


Fig. 18 Comparison between the concentrations of methane and carbon dioxide surrounding (a) Matuail landfill and (b) Aminbazar landfill obtained by field measurements and AER-

MOD View air pollution dispersion modeling based on four simulations of LandGEM V-3.02 model

by field measurements and AERMOD View air pollution dispersion modeling based on four simulations (CAAC, IC, IW, SP-1) of LandGEM V-3.02 model revealed that field measurement findings obtained through measuring by Haz-Scanner-TM model HIM-600 provided higher concentrations of both CH₄ and CO₂ as compared to the model outputs. This might be due to the reason that AERMOD View model only considered landfills as the source of CH₄ and CO₂ emissions. However, in real scenario decomposition of organic waste in landfill sites along with other sources such as transportation and industrial activities (Elmi et al., 2021) can combinedly emit gases like CH₄ and CO₂ resulting in the higher concentrations of these gases obtained by field measurements. Moreover, AERMOD View model assumed the background concentration of CH₄ and CO₂ to be zero, whereas in real scenario these gases are already present in the ambient atmosphere due to contribution from multiple other sources as mentioned earlier. This assumption might also lead to lower modeled concentrations of the gases compared to field measurements. In their study, Toha & Rahman, 2023 also found models to be overestimating methane emissions compared to in-situ measurements. The comparative analysis also revealed that dispersion modeling based on emission data from LandGEM IW was the closest to the concentrations of CH₄ and CO₂ obtained by field measurements. Therefore, AERMOD View model can be used based on emission data from the IW simulation of LandGEM V-3.02 model to accurately assess the

dispersion pattern of the emitted gases from Matuail and Aminbazar landfills. This outcome aligns well with the climatic characteristics of Dhaka. The regional dryness index for Dhaka was determined to be 79.85 ($I > 35$), which categorized the area as having intensely humid conditions (Table 1). The IW (Inventory Wet) simulation aims to represent landfills operating under wet climatic conditions, where higher moisture content in waste enhances anaerobic decomposition and production of landfill gas. Since the Matuail and Aminbazar landfills receive predominantly organic waste (67.65% of MSW) (Toha & Rahman, 2023) along with high humidity and rainfall, dispersion modeling based on emission data from LandGEM IW simulation is the most appropriate for Dhaka's conditions.

5 Limitations of the Study

One of the limitations of the study was estimation of emission rates of landfill gases using the LandGEM model. LandGEM is based on default values for input parameters such as methane generation rate (K), and potential methane generation capacity (L_0). Since these default values may not fully represent the actual waste composition and environmental conditions of Dhaka's landfills, the modeled outputs could differ from real emission patterns. Moreover, the dispersion analysis was conducted using meteorological data from only a single year (2023) following previous

studies (Elmi et al., 2021; Talaiekhosani & Nasiri, 2016; Talaiekhosani et al., 2016, 2018a, b; Wijaya et al., 2021), which may not sufficiently represent long-term variability in wind patterns and atmospheric conditions. Moreover, solar radiation and precipitation data could not be included due to unavailability. Field measurements were also limited to three sampling points per landfill, with one-hour monitoring at one-minute intervals following Toha and Rahman (2023) who conducted field measurements for 30 min at 5-min intervals. This design may not capture the broader spatial and temporal variability in gas concentrations, which may contribute to discrepancies when compared with model outputs. Additionally, the study did not account for seasonal variations, even though dispersion dynamics can vary considerably across seasons. The dispersion modeling itself involved simplifying assumptions, such as zero background concentrations of CH₄, CO₂, and NMOCs, and negligible contributions from other emission sources such as transportation and industrial activities, which may not reflect actual conditions. These factors likely contributed to the observed bias between modeled outputs and field measurements, where simulations tended to underestimate gas concentrations.

6 Conclusion

Landfills are currently the exclusive means of disposing municipal solid waste (MSW) in Dhaka, Bangladesh and are expected to continue being the primary technique in the years to come. Different operations of landfill sites emit gaseous pollutants including greenhouse gases (GHGs) like methane (CH₄), carbon dioxide (CO₂), and other non-methane organic compounds (NMOCs), that represent a threat to both the environment and human life. In this study, emission rates of CH₄, CO₂, and NMOCs from Matuail and Aminbazar landfill sites were estimated by using four simulations (CAAC, IC, IW, and SP-1) of LandGEM model and the atmospheric dispersion pattern of the emitted gases from landfills was identified using AERMOD View Software. Field measurements were also carried out to compare with the model outputs. The maximum concentrations of CH₄, CO₂, and NMOCs were found near the landfill sites. However, the maximum concentrations of the emitted gases did not exceed the permissible limits indicating no health

threats from the emitted gases to the landfill workers. Among the four simulations of LandGEM model, dispersion modeling based on emission data from IW provided findings that were closest to the field measurements.

This study extends its relevance beyond the specific site, holding broader implications for the region as a whole. The initial validated estimates of landfill gas emissions and their dispersion in Dhaka create an essential baseline for evaluating and controlling air pollution stemming from waste. The findings emphasize the unequal exposure risks faced by surrounding communities, reinforcing the necessity of integrating landfill emissions into frameworks for occupational safety, environmental health, and urban planning. The study also emphasizes the importance of incorporating landfill gas management into Bangladesh's comprehensive environmental and climate strategies from a policy standpoint. Methane emissions from landfills have a direct impact on national commitments related to the Paris Agreement and the Global Methane Pledge, which aims to achieve a minimum reduction of 30% in global methane emissions from 2020 levels by 2030. The baseline data produced in this study can serve as a foundation for policymakers to establish emission benchmarks, develop monitoring systems, and implement targeted interventions aimed at decreasing greenhouse gas emissions associated with landfills. The findings offer practical insights for enhancing landfill operations in Dhaka and also align with the broader goals of sustainable waste management, air quality enhancement, and climate change mitigation at the national level. Enhancing landfill gas management, in conjunction with comprehensive waste reduction and resource recovery approaches, will be vital for safeguarding public health and promoting sustainable urban development in Bangladesh.

The study recommends that it is necessary to properly design and operate an efficient landfill gas extraction facility in both Matuail and Aminbazar landfills. The gas extraction facility should include vertical extraction wells in deeper waste zones, supplemented by horizontal collectors in active disposal areas, to maximize gas recovery across the landfill body. This will help to reduce emissions of greenhouse gases like methane (CH₄) and carbon dioxide (CO₂) and other gaseous pollutants such as non-methane organic compounds (NMOCs), carbon monoxide (CO), oxides of sulfur (SO_x), oxides of nitrogen (NO_x)

etc. into the atmosphere from these two landfill sites and thereby tackling air pollution from landfill sites in Dhaka. The information obtained in this study can offer essential insights for the effective design and operation of landfill gas extraction systems in both Matuail and Aminbazar landfills. The study also recommends establishing national standards for methane capture efficiency and allowable emission levels, ensuring landfill gas collection and utilization for all active and future landfill sites, introducing reporting requirements for landfill operators, including regular emissions inventories and monitoring data, and creating financial and institutional incentives for public–private partnerships in landfill gas-to-energy projects. By implementing these measures, air pollution from landfill gas emissions in Dhaka can be reduced, while simultaneously transforming landfill gas into a valuable renewable energy resource.

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Declarations

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